

Answers & Explanations

the catapult papers



Warm-Ups 1-10

2026 edition

Notes for parents

Answers booklets in The Catapult Papers include explanations for every answer.

Each explanation is brief, however, as they are limited to two lines per question. You should nevertheless discover a broad range of helpful insights and tips that could, in turn, be used to explain mistakes to your child. Should you find yourself requiring further explanations for any of the questions in this series, please do not hesitate to contact me (Trevor) at info@catapult-tuition.com.

When reading the explanations throughout this series, you will come across a range of strategies. For example, when it comes to spellings challenges, I will occasionally point out regular spelling rules and patterns that can be transferred across many other words. That said, we are also aware that the English language has many inconsistencies and confusing spellings; this means that it is necessary, on occasion, to be more creative (even childlike) in creating 'memory hooks' for certain words.

You will also discover that, for certain Maths questions, I would encourage children to test out each option (this is one of the benefits of multiple-choice questions) but, on other occasions, I will point out fast 'tricks' that could help students come to the answers quickly. There is rarely one 'correct' way to solve a Maths question. It is up to you to discern which approaches work best for your child.

It should also be emphasised strongly that **this booklet is written for adults**. While you can hand this booklet over to your child for self-marking purposes, please do not encourage them to read the explanations. True, there will be many explanations that children will readily understand but there are also many technical terms used that are intended for adults (particularly in the English sections).

Finally, please do not allow your child (or yourself!) to become disheartened by 'low' scores, especially in early tests. As you will no doubt discover, the Transfer Tests often include questions that we adults would answer incorrectly. The idea of practice is to slowly but surely grow in knowledge, skills and understanding. Indeed, by the end of the Transfer Test season, parents who are new to the process often find they have learned new and interesting things. That doesn't mean these months are going to be fun... but we should take what we can get!

Warm-Up 1 English Answers

1	B	A question mark is needed, rather than a comma. There must always be punctuation before a closing speech mark. In most cases, this will be a comma. However, whenever a question is asked, there must always be a question mark.
2	A	The word <i>wasn't</i> requires an apostrophe for contraction . This is because it is a combination of two words (<i>was</i> & <i>not</i>) but with a letter missing (the letter <i>o</i>). The missing letter (or letters) must be replaced by an apostrophe.
3	D	A full stop is missing at the end of the sentence. Also note the correctly used possessive apostrophe in the word <i>everyone's</i> (the dinner belonged to <u>everyone</u>).
4	N	There are no mistakes. Exclamation marks are used to suggest strong emotion (not necessarily shouting). We must assume that Mrs Thorpe spoke both sentences with a sense of urgency or emotion, as there is no evidence that she didn't.
5	C	<i>Newcastle</i> is a proper noun , as it is the name of a town, and must therefore begin with a capital letter .

6	D	<i>Everything that Louisa wrote was presented in the most beautiful handwriting.</i> The simple past tense of <i>write</i> is used, which is wrote . The word <i>written</i> is a past participle, and will usually only appear after words like <i>have</i> / <i>has</i> / <i>had</i> .
7	A	Instead of <i>complaining about it, why don't you just do something?</i> Reading the second word in the sentence is very important: of . From the five options, only the word instead can be sensibly followed by <i>of</i> .
8	D	<i>To be honest, this has been a complete waste of my time.</i> Some children have difficulty in distinguishing <i>of</i> from <i>off</i> . It may help to focus on their pronunciation within a sentence. <i>Off</i> must always emphasise the 'f' sound, whereas <i>of</i> usually sounds like 'uv'.
9	C	<i>When I arrived at the hotel yesterday, I could immediately see that it was all I'd ever wanted.</i> The sentence is in the past tense. We are looking for the contraction of the words <i>I had</i> (...everything <i>I had</i> ever wanted).
10	B	<i>The writing on the wall could have meant anything; it might as well have been in a foreign language.</i> The phrase is could have . When spoken quickly, it does sound like 'could of' (particularly when we use the contraction <i>could've</i>).

11	C	The spelling should be immediate . The spelling contains the word 'media' (perhaps children have seen this word when discussing 'social media'). They could also check question 9!
12	D	The spelling should be mysteries . The noun <i>mystery</i> finishes with a single y. There is no vowel before it. As such, the y must be converted into the common <i>-ies</i> plural ending (e.g. <i>baby/babies</i> ; <i>party/parties</i>).
13	C	The spelling should be opening . There is no double n. This is because the word <i>open</i> is emphasised on its first syllable. Words that have a vowel-consonant ending are often doubled (e.g. <i>begin/beginning</i>) but only when that particular syllable is stressed.
14	C	The spelling should be respectful . The <i>-ful</i> suffix does not use a double L, even though it is associated with the word <i>full</i> . Other examples are <i>beautiful</i> , <i>awful</i> and <i>wonderful</i> .
15	N	There are no mistakes. As the word <i>journey</i> finishes with a 'vowel + y' combination, we do not use the typical <i>-ies</i> ending. <i>Grateful</i> is often misspelled as 'greatful' but has nothing to do with the adjective 'great' (rather, it is connected to <i>gratitude</i>).

16	C	The purpose of the first paragraph is to introduce us to the main character (Finn McCool) as well as set the scene for the main event in the story (<i>the stunning coast of County Antrim</i>).
17	B	Although quicksilver thinking may seem an unusual phrase to children, they should hopefully be able to connect 'swift' to 'quick'. Those who chose 'sheer brilliance' (line 26) should check the relevant sentence: those words are not about Finn McCool.
18	E	Line 19 contains the words that are opposite in meaning to each other: <i>reality</i> and <i>fantasy</i> .
19	A	The clue is on lines 23-24: <i>more and more flickers of uncertainty danced across his face.</i> 'Have I met my match?' Benandonner was having second thoughts about his chances of success.
20	E	The word <i>ingenious</i> is describing the proper noun <i>Oonagh</i> . It is therefore an adjective .
21	C	<i>Offspring</i> is another word for 'baby' (line 28). It is the 'baby' that is monumental in size. Benandonner was thinking that if the 'baby' was so huge, then the father must be even more huge.
22	A	Everything written before the possessive apostrophe is the name of the owner/possessor. In this case, the word <i>Giant</i> appears before the apostrophe in <i>Giant's Causeway</i> . This means that the causeway belongs to one giant.

23	turned tail	The phrase turned tail (line 31) is an idiom that means 'to suddenly run away or retreat'. In the animal kingdom, when an animal feels threatened, it may 'turn its tail' and run away.
24	chronicle	The word chronicle (line 33) is a written record of a historical event.
25	storyteller	This word is on line 4. A compound word is a combination of two or more words into one, without the use of punctuation marks (e.g. hyphens or apostrophes). <i>story</i> + <i>teller</i> = <i>storyteller</i>
26	14	<i>The line number must be written.</i> The simile is ... <i>like a lightning bolt</i> . Similes are phrases that compare things because of something they have in common. They use the words <i>as</i> or <i>like</i> .
27	spouse	Although children may not yet be familiar with the word spouse , they should see that Oonagh is Finn's <i>spouse</i> (line 26), and that Finn is her <i>husband</i> (line 27).
28	verb	These are all action (or 'doing') words within their respective sentences.

Warm-Up 1 Maths Answers

29	A	Chloe reads 80 pages in total from Monday to Friday (16x5), and adds another 22 to this total on Saturday, making 102. Subtract this from the book's page-count of 127 to get Sunday's amount: 25 pages .
30	B	All three angles in any triangle add to make 180°. The angles of 66° and 59° add to make 125°. Subtract this from 180° to get 55° .
31	C	900ml of fruit juice are required to make <u>ten</u> lollies. We only need to work out the amount needed for <u>two</u> . To do this, we can divide by ten then multiply by two, or we can simply divide by five. $900\text{ml} \div 5 = \mathbf{180\text{ml}}$
32	B	The central zone of the Venn Diagram is for numbers that are <u>prime</u> as well as <u>factors of 12</u> (this means numbers that fit perfectly into 12). 1 is not a prime number. 2 & 3 are both prime <u>and</u> factors of 12, so the answer is two numbers .
33	C	Do <u>not</u> divide 36 by 4. The area of a square is calculated by multiplying length by width. As this is a square, these two measurements must be the same: 6cm x 6cm. The perimeter is therefore 24cm (6cm + 6cm + 6cm + 6cm).
34	A	25 tens = <u>250</u> . 25 thousands = <u>25000</u> . 25 units = <u>25</u> . $250 + 25000 + 25 = \mathbf{25275}$
35	B	If there are 12 children at the party, and Mrs Thorpe wants each child to have three slices of pizza, there needs to be 36 slices of pizza altogether. There are six pizzas. $6 \times \square = 36$. There must be six slices in each pizza.
36	D	90 minutes is 1 hour and 30 minutes. Add one hour to 6.45pm to reach 7.45pm. Split the 30 minutes into two 15-minute segments (to reach 8pm, then 8.15pm). Be careful to not choose the option that finishes with <u>am</u> .
37	B	The three bottles of water cost <u>£2.55</u> ($85\text{p} \times 3$). The two energy bars cost <u>£2.98</u> ($\text{£}1.49 \times 2$). Add these amounts together to get £5.53 .
38	E	135° is the same as one right angle (90°) plus another 45° movement. Moving clockwise from north, one right angle brings us to east, then another 45° turn brings us to southeast .
39	D	12.00pm to 3.40pm is somewhere between 3 hours and 4 hours. It is within the range of <u>180 - 239</u> minutes. Locate this part of the table on the <u>North St</u> row. The charge is <u>£6.25</u> . Subtract this from £10 to get £3.75 .
40	D	Year 1 won <u>17</u> prizes. Year 4 won <u>8</u> . The difference between these numbers is 9 .
41	A	A square has four lines of symmetry . A parallelogram has none. An equilateral triangle has three. A rectangle has two. A rhombus also has two.
42	C	All of the statements describe perpendicular paths (crossing at right angles to each other) apart from Statement C . These pairs of characters are walking in the same direction, and would not cross. They therefore walk along parallel paths.
43	E	There are 15 minutes in one quarter of an hour. There are 60 seconds in a minute. Multiply 15 by 60 to discover the total number of seconds: 900 . This calculation can be performed faster by first doing 15×6 , then multiplying the answer by ten.
44	A	The tenths place is immediately after the decimal point. This is where the digit 9 sits.
45	E	If Jonah's pace is 6km/h then it takes him <u>half an hour</u> to walk 3km (because this is half of 6km). He leaves home at 8.15am, so will arrive at school at <u>8.45am</u> . Saoirse also arrives at this time (8.35am + 10 minutes).
46	A	Each longer line on the scale is a multiple of ten. As it takes five 'steps' to reach each of these, then <u>each smaller interval must represent 2°C</u> . Count back three of these steps from 0°C to reach -6°C .
47	C	The sequence of triangular numbers uses the following adding pattern: 1 (+2) 3 (+3) 6 (+4) 10 (+5) 15 (+6) 21 (+7) 28 (+8) 36 (+9) 45 (+10) 55 .
48	B	1300g must first be divided by five to discover how much the dog eats in one day (260g), then multiplied by seven to discover how much the dog eats in one week (1820g). This is converted into kilograms (divide by 1000): 1.82kg .
49	C	Write $\frac{7}{9}$ beside each of the five options. Only one of the options will contain a numerator and denominator that are equally greater than the 7 and 9. It is $\frac{42}{54}$. This is because 42 is <i>six times greater</i> than 7, and 54 is <i>six times greater</i> than 9.
50	B	The pattern is increasing in square numbers. Shape 1 has four squares (2x2), Shape 2 has nine squares (3x3), Shape 3 has sixteen squares (4x4). Shape 4 must therefore have 25 squares (5x5).

51	scalene	The 'slanted' side (known as the hypotenuse) cannot measure the same as the base. It has to be longer than 15cm. This means all three sides have differing lengths, making it a scalene triangle.
52	1.141	<u>11.41</u> is the largest of the numbers. When divided by 10, this makes 1.141 .
53	6 cm	The perimeter of the square must be 32cm (8cm + 8cm + 8cm + 8cm). The rectangle's perimeter is therefore 20cm. Subtract two 4cm lengths from this (12cm) then divide by two. The answer is 6cm .
54	45 edges	All of the edges are visible in the image because the shape is transparent: kids can count the lines if they can't remember the facts. The prism has 9 edges. Five of these would make 45 edges .
55	(8.5)	First locate the number on the lower axis (the x axis). The mark sits above the number 8 . Then locate the number on the side axis (the y axis). The mark sits across from the number 5 .
56	10744	To calculate the number of attendees on the second night, subtract 480 from 5612. This makes 5132. Add this to 5612 to make 10744 .

Warm-Up 2 English Answers

1	B	The word <i>it's</i> should have an apostrophe for contraction : it is short for 'it is'. Please note: the comma is not a mistake. It is acceptable, though often not necessary, to place a comma before conjunctions such as 'and' and 'but'.
2	N	There are no punctuation mistakes in this sentence. The use of brackets is helpful in explaining who Bonnie is. Using a pair of commas or dashes instead can often create the same effect.
3	C	The words <i>dogs</i> should not have an apostrophe . It is not a contraction of 'dog is' (or 'dog has'). A possessive apostrophe cannot be used either, as the dog/dogs are not said to possess anything.
4	D	A question mark is needed at the end of the sentence: ' <i>...but where was Hannah?</i> ' Please note: the use of a comma before the word 'but' is not a mistake. See question 1.
5	B	A second sentence has begun; the word <i>The</i> needs to be capitalised .

6	E	<i>I won't be able to attend the meeting as I have a doctor's appointment.</i> The most sensible word to use in this context is as . It carries the same meaning as 'because'.
7	A	<i>'Please help yourself to some coffee while we fix it.'</i> For options B to E we would not use reflexive pronouns; we would use <i>him, me, her</i> or <i>us</i> instead.
8	D	<i>It was, without doubt, the most beautiful sunset that they <u>had</u> ever seen.</i> The key word is <i>had</i> : ' <i>...<u>had</u> ever <u>seen</u>.</i> ' Even though the simple past tense of 'see' is 'saw', we must use seen in this sentence because of the preceding verb <i>had</i> . It is a past participle.
9	A	<i>He was caught stealing some rubbers from the stationery cupboard.</i> This is the simple past tense, as well as the past participle, of <i>catch</i> . He was caught [<i>in the act of</i>] stealing...
10	E	<i>Harvey's blankets and clothes were spread all across the bedroom floor.</i> The simple past tense, as well as past participle, of <i>spread</i> is still spread . There is no such word as <i>spreaded</i> or the other options, apart from <i>sprayed</i> (the past tense of 'spray').

11	A	The spelling should be believed . This word follows the famous 'i before e, except after c' rule (which only applies to words in which the two vowels make the long 'ee' sound). We can also see that the word 'lie' is in the middle of believe .
12	D	The spelling should be situation . The suffix <i>-tion</i> is the most common spelling for the 'shun' sound.
13	C	The spelling should be deserted . A double s spelling refers to the final course of a meal.
14	N	There are no spelling mistakes. Achievement follows the <i>i before e, except after c</i> rule (see question 11). The final e of <i>achieve</i> is not dropped as the suffix <i>-ment</i> does not begin with a vowel.
15	D	The spelling should be focus . Even when we add the <i>-ed</i> or <i>-ing</i> endings, the s should not be doubled. This is because we stress the first syllable (<i>fō-cus</i>), and not the syllable that contains the <i>-us</i> ending.

16	C	The passage builds up to the conclusion of Messi's career: winning the World Cup. Lines 35-36 gives us a very strong clue to the answer: <i>Argentina triumphed in the penalty shoot-out, providing a fitting conclusion to Messi's career.</i>
17	B	See lines 6-7: <i>He joined the local youth team, Newell's Old Boys, at the age of eight.</i>
18	A	Lines 14-15 provide the biggest clue: <i>...dribbling skills...lightning-quick movement...</i> Although we are not told in the passage about Messi's height, his nickname (<i>The Flea</i> : line 15) suggests that he must have been small.
19	D	Find each word in the passage and swap them with <i>leading lights</i> (the plural ending is a clue!). Only option D seems sensible. Additionally, the spelling of luminaries suggests a connection to <i>lights</i> (e.g. <i>illuminate, luminous</i>).
20	E	Lines 23-25 provide the answer: his time at PSG was <i>underwhelming in relation to his previous achievements</i> .
21	A	From line 27 onwards, we learn that Messi had never won the World Cup. After we read of him finally winning it, we learn that he was able to <i>perform on the grandest stage</i> . He won the trophy (option C) while performing on the <i>stage</i> (option A).
22	E	<i>Chronological</i> refers to the arrangement of events, dates, or historical items in the order in which they occurred, from the earliest to the most recent. This is exactly how the passage is written: it is a chronological report .

23	evident	It is important to swap words around to see if they work. The word is on line 6: <i>Messi's love for the sport was evident [clear] from a young age.</i>
24	Ballon d'Or	See line 18: <i>...he won six Ballon d'Or awards...</i>
25	previous achievements	We should swap two-word phrases on these lines with the words <i>past accomplishments</i> . We can tell that we are probably looking for a plural, so this narrows our options. See line 24.
26	The Move to Paris	The information about money problems is on line 22 (<i>financial difficulties</i>). Children must find the subheading for this section and copy it. It is on line 20.
27	Argentina	Although Messi played most of his career in Spain (with Barcelona), his first team was based in his home country of Argentina (Rosario is a <i>city</i>). The clue is on line 6: they were a <i>local</i> team.
28	adverb	Children may already be aware that most adverbs have an -ly ending (although they should be careful, as words like 'lovely' and 'silly' are typically adjectives).

Warm-Up 2 Maths Answers

29	B	25% is the same as one-quarter. Divide 280 by four to find the answer: 70 .
30	E	The fourth corner of the rectangle will be at (8,3) . We must read along the lower (horizontal) axis first to get the first number (8), then rise up parallel to the vertical axis to line up with the second number (3).
31	C	$1000 \times 5 = 5000$. However, this means 5000p (not £5000). Insert a decimal point before the final two zeros to convert this amount into pounds: £50.00 .
32	D	In an <u>isosceles</u> triangle, two side lengths are the same. In this case, both the lower angles are 59° . All three angles in a triangle must add to make 180° . Angle A must be 62° . $180^\circ - (59^\circ \times 2)$.
33	C	There are 144 chairs in total, so our answer cannot be greater than 144. It is solved by division: $144 \div 8 = \mathbf{18}$.
34	D	There are 1000 metres in one kilometre. This means that the number of metres is 1000 times greater than the number of kilometres. We must multiply 49.5 by 1000: 49500m .
35	C	The digits in <u>any</u> multiple of three will add to make another multiple of three (for example, the digits in 24 add to make six; the digits in 36 add to make nine). Both 51 and 57 follow this pattern (six & twelve), so the answer is two numbers .
36	A	7% is the same as seven-hundredths. When written as a decimal, the digit 7 must be in the hundredths place. The answer is 0.07 .
37	E	The volume of a cube is calculated by multiplying its length by its height, and then by its depth. This cube's volume is therefore calculated by multiplying 9 by 9 (81), then by 9 again. The answer is 729cm^3 .
38	C	Rather than repeatedly adding £1.75, children should multiply by ten to reach £17.50. This leaves only £2.50, which is enough for just one more bagel, making the final cost £19.25 . The change from £20 would be 75p .
39	B	25 is a square number, and is odd, so goes in box 1. 36 is a square number, but is not odd, so goes in box 2. 42 is not a square number, but is a multiple of 6, so goes in box 3. 122 is neither a square number nor a multiple of 6, so goes in box 4.
40	C	No long multiplication needs to be carried out. The answer to our question is <u>half</u> the answer to the previous multiplication. This is because the 42 has been halved to become 21. The answer is 336 .
41	E	The horizontal sections of the line show when the car was not travelling (the distance did not increase, so the car must have been stationary). As each square on the grid represents 30 minutes, the car did not travel during 120 minutes of the journey.
42	B	$288 \div 9 = 32$ $32 + 26 = \mathbf{58}$
43	D	The information on the right of the table tells us that the bus leaves Durham every 60 minutes after 0715. Colleen will have to catch the bus at 0815. Each time on the table increases by one hour. She will arrive at Houghton-le-Spring at 0844 .
44	B	Although rhombuses are being used, they are arranged in <u>square number</u> patterns: 1x1, 2x2 and 3x3. Pattern 4 will be a 4x4 pattern, and therefore have 16 rhombuses .
45	A	The length of time it took Joel to complete the jigsaw was 1 hour 34 minutes. This converts to 94 minutes .
46	E	There are 30° between each number on a clock face ($360^\circ \div 12$). The reflex angle at 10 o'clock is the clockwise angle from the minute hand to the hour hand. It spans ten numbers, so the angle must be 300° ($30^\circ \times 10$).
47	B	$21.975 \times 1000 = 21975$. The digit in the tens place is 7 .
48	C	Test each option out to see if a consistent pattern appears. The gap from 47 to 33 is 14, which suggests a 'subtract 14' pattern. This would place a 19 between the 33 and 5. The first missing number would be 61 .
49	E	It takes four intervals to get from one litre to the next, meaning that each interval is worth 250ml ($1000\text{ml} \div 4$). There are 2750ml of juice in the jug.
50	C	There are two hexagonal faces on the prism (front and back). This means that there needs to be six rectangular faces (one for each side of a hexagon). There are 8 faces in total.

51	20 pretzels	Each image represents 20 pretzels. The pictogram shows 4 images for Saturday, which means <u>80</u> pretzels. Thursday shows 3 images, which means <u>60</u> pretzels. The difference is 20 .
52	1400 roses	40 must be multiplied by 35 to get the answer. The fastest way to do this is to multiply 35 by 4 (140), then multiply the answer by ten: 1400 .
53	300°	The full circle would have measured 360° . The size of the angle that has been removed is the top of an equilateral triangle, which is <u>always</u> 60° per angle. $360^\circ - 60^\circ = \mathbf{300^\circ}$.
54	135 cm^3	The volume of a cuboid is calculated by multiplying length by height by depth. $9\text{cm} \times 5\text{cm} \times 3\text{cm} = \mathbf{135\text{cm}^3}$.
55	23:20	00:50 means 12.50am. 60 minutes before this is 11.50pm. Then subtract another 30 minutes to reach 11.20pm. As the display is showing 24-hour time, convert this to 23:20 .
56	91 km	The Enniskillen to Newry distance is <u>106km</u> (it is across from Enniskillen and below Newry). The Belfast to Lisburn distance is <u>15km</u> (below the vertical Lisburn). The difference is 91km .

Warm-Up 3 English Answers

1	C	A comma is needed between <i>dance</i> and <i>act</i> . The exclamation mark is fine as it can suggest strong emotion in the tone.
2	D	There is no full stop at the end of the sentence . Children should not over-concern themselves with the appearance of semi-colons (or colons). Any semi-colon mistakes in test papers are likely to be obvious.
3	A	<i>Wednesdays</i> should not have an apostrophe . It is not a contraction of 'Wednesday is' (or 'Wednesday has'). Neither is there a need for a possessive apostrophe, as nothing is stated as belonging to Wednesday.
4	C	<i>Brother's</i> should have a possessive apostrophe . The <i>achievements</i> belong to the brother . The apostrophe goes after the word <i>brother</i> , then an s is added afterwards.
5	A	An opening speech mark is needed at the very beginning of the sentence. Dad is talking.

6	E	'There is no name at the top of the page, so I can't tell whose work it is.' Whose means 'belonging to whom'. The teacher wants to know whom the work belongs to. The word <i>who</i> is a pronoun, so does not get a possessive apostrophe (e.g. my, your, his).
7	B	It would be up to <u>either</u> Mia or Leo to break the bad news to everyone. The use of the word <i>either</i> is a clue, as it is usually paired with the corresponding or . By contrast, <i>neither</i> would be paired with <i>nor</i> .
8	C	It was <u>a</u> remarkably hot day in the city of Houston. An adverb is needed to describe the adjective <i>hot</i> (yes, adverbs can also describe adjectives!). It cannot be <i>incredibly</i> , which begins with a vowel and would have to be preceded by the word 'an'.
9	D	Charles <i>stubbornly insisted upon taking the front passenger seat of the car</i> . We either 'insist on' or 'insist upon' something. The only one of these prepositions available among the options is upon .
10	A	I'll meet you where the river bends and the trees start to thin. <i>At</i> would only have worked if the sentence ended at the word <i>river</i> . But, as we are locating a specific place somewhere along the river, we need to use <i>where</i> .

11	A	The spelling should be requirement . The letter e is not dropped from the word 'require'. This is because the <i>-ment</i> suffix does not begin with a vowel.
12	N	There are no spelling mistakes. In British English, the metric words (e.g. <i>metre</i> , <i>centimetre</i>) have an <i>-re</i> ending. There is also a double L in <i>millimetres</i> , as this prefix is connected to words associated with 1000 (e.g. one <i>million</i> is a thousand thousand).
13	B	The spelling should be irresponsible . There is a double r. This is because the prefix <i>ir-</i> has been added to the word <i>responsible</i> . One r belongs to the prefix, one r belongs to the main word (other examples are 'irregular' and 'irrelevant').
14	A	The spelling should be would (a homophone of <i>wood</i>).
15	C	The spelling should be gloriously . Although the noun 'glory' ends with the letter y, this has to be exchanged for an i when adding a suffix to it.

16	A	The poem does not have a rhyming pattern . Although <i>embrace</i> and <i>haste</i> (lines 41 & 42) could be said to rhyme, there is no <u>pattern</u> of rhyme throughout the whole poem.
17	D	The poem has a peaceful mood. The son is reflecting upon the demands of school, and realising that the beauty of nature can help him forget this for a while, putting his troubles away.
18	C	Lines 5-6 contain examples of alliteration, which is the repeated use of the same starting sounds in words (the opposite of rhyme): <i>...the forest floor, a field of fallen foliage... scrunching softly beneath our synchronised steps...</i>
19	E	Lines 17-18 provide the clue: the father and son have <i>no scheduled path</i> , and the word <i>wandering</i> is used. They do not mind which path they take. They are simply going for a walk.
20	E	Swapping each option for the word <i>wild</i> will help. For example, it rules out option A (<i>...the forest, a wild in our shared journey...</i> makes no sense). The answer is untamed : <i>...the silent poetry of the untamed [the wild]...</i>
21	B	The whole point of the poem is to show that the son must forget his concerns , and enjoy nature. Lines 29-32 and 41-44 provide clues for this. Furthermore, the father's purposely vague answer in line 28 hints that the son needs to forget about school.
22	E	<i>Beneath</i> is a preposition in this line. These are 'position words' that show where one thing is in relation to something else. In this case, the father and son are <i>beneath</i> Sherwood's ancient umbrella (a metaphor for 'shade' or 'canopy').

23	sanctuary	A sanctuary (line 4) is a <i>safe place</i> . Perhaps children have heard of 'animal sanctuaries' and can make the connection.
24	31	The simile is <i>worries disperse like morning mist</i> . Similes compare one thing (the son's worries) to a different thing that has something in common (morning mist). Similes use the words <i>as</i> or <i>like</i> .
25	absorb his concerns	This phrase is on line 27. <i>Soaking up</i> is connected to the word absorb , while <i>worries</i> can mean the same as concerns .
26	(glimmers of) sunlight	The answer is found on lines 9-10. The sunlight creates the mosaic. Children don't have to write all three words to get marked correctly; the words <i>glimmers</i> or <i>sunlight</i> by themselves are correct.
27	happier	Spelling must be perfect. This question is designed to see if children know to exchange the y at the end of <i>happy</i> for the letter i before adding the <i>-er</i> suffix.
28	noun	Within the poem, these words are all <u>things</u> , which makes them nouns. Words like <i>steps</i> and <i>whispers</i> are commonly used as verbs, but not on this occasion.

Warm-Up 3 Maths Answers

29	B	Take one hour from 20:05 to reach 19:05. Split the 40 minutes into 5 minutes and 35 minutes. 19:05 subtract 5 minutes is 19:00. 19:00 subtract 35 minutes is 18:25. This is the same as 6.25pm .
30	B	Point B is the only reflection of a corner of the triangle. Count across horizontally from the top corner of the triangle to the mirror line (four squares), then count four squares further to determine the reflection point.
31	E	We must first count how many shapes there were altogether (12 circles, 7 semi-circles, 10 squares, 9 rectangles and 10 triangles). There are 48 shapes. 12 is one-quarter of 48, which matches 25% .
32	E	$24 \times 6 = 144$
33	C	First, change five minutes into seconds: 300 seconds (60x5). Find nine-tenths of 300 by dividing by ten then multiplying by nine. $300 \div 10 = 30 \dots 30 \times 9 = 270$
34	A	You cannot swap the numbers when dividing and expect the same answer. The question is <u>not</u> asking for 100 divided by 4. We must make 4 one-hundred times smaller. 'Shift' the digit 4 two places to the right: 0.04 .
35	D	'Consecutive' means 'next to each other' or 'in uninterrupted order'. The two numbers must therefore be 11 and 12. These add to make 23 (the sum), and multiply together to make 132 (the product).
36	B	Option A: $8 \times 25g = 200g$. Option B: $4 \times 65g = 260g$. Option C: $(4 \times 25g) + (2 \times 65g) = 230g$. Option D: $25g + (3 \times 65g) = 220g$. Option E: $(5 \times 25g) + 65g = 190g$.
37	E	24 has eight factors (numbers that fit perfectly into it): 1 and 24, 2 and 12, 3 and 8, 4 and 6.
38	A	A kite can only have one line of symmetry (for the one in the image, it would be a vertical line down through the centre). They may contain a right angle. There are never any parallel sides.
39	A	45° is half of a right angle, which means 'half of a quarter' of the circle. This means one-eighth of the total. $\frac{1}{8}$ of 72 people is 9 people .
40	D	A quarter-turn is 90° (one quarter of a full turn). Poppy would move <u>clockwise</u> from facing east to facing south .
41	C	The area of the entire garden is 112m ² (14x8) but it is not entirely covered with grass. The area of the path must be subtracted. The area of the path is 14m ² (14x1), meaning that the area covered with grass is 98m² (112 - 14).
42	E	Option B is impossible. Fewer than half the numbers on a dice are square numbers (1 & 4) so option A is unlikely. Half of the numbers are odd: option E has an even chance. Five of the numbers are factors of 12, so option C is likely. Option D is certain.
43	C	0.06 means six hundredths. One hundredth of a kilometre (1000m) is 10 metres, so six of these would make 60m .
44	D	The hundredths place is the second digit to the right of the decimal point. The digit 8 is in the hundredths place.
45	A	The bottom-right angle in the isosceles has to be 50°. Angle A must therefore be <u>80°</u> (180° - 100°). A quadrilateral's angles add to make 360°, so add the three known angles (285°) then subtract from 360° to get Angle B: <u>75°</u> . The difference is 5° .
46	C	The number of coloured squares in each pattern will be <u>one less than the square of an odd number</u> . Pattern 1 has one less than 3 ² . Pattern 2 has one less than 5 ² . Pattern 3 has one less than 7 ² . This means Pattern 4 will have one less than 9 ² , which is 80 .
47	D	$8 \times 22 = 176$ $176 - 2.2 = 173.8$
48	A	Each option has 18 as a whole number, so find the number with the largest tenths digit. 18.6 has a 6 in the tenths place, which is greater than what all the other options have in this decimal position. It is therefore the largest number in the list.
49	B	$\frac{3}{4}$ of 16km is 12km ($16 \div 4$ then $\times 3$).
50	A	The <i>sum</i> is the answer to an addition, so the question is really $979 + \square = 2861$. Perform a subtraction to find the missing number: $2861 - 979 = 1882$.

51	£16	The <i>range</i> is the difference between the largest and smallest values in a list. The largest amount of money is £48, and the smallest is £32. $£48 - £32 = \mathbf{£16}$.
52	£195	There are <u>three months</u> in one quarter of a year (12 months $\div$ 4). $£65 \times 3 = \mathbf{£195}$.
53	10 minutes	Jamie arrives at the station at 2.55pm, which is 1455. Read across the table from 'Coventry' to see the departure times. He can catch the 1505 train, which comes in ten minutes .
54	75	$a = 43$. $b = 23$. $c = 9$ $43 + 23 + 9 = \mathbf{75}$
55	49 views	Wednesday had 18 views. Thursday had 10 views. Friday had 21 views. The sum of these numbers is 49 .
56	16 pupils	$\frac{1}{3}$ of 36 pupils is 12 ($36 \div 3$). $\frac{2}{3}$ of 36 pupils is 8 ($36 \div 9$ then $\times 2$). This makes 20 pupils, leaving 16 pupils to walk to school.

Warm-Up 4 English Answers

1	D	The sentence is missing a full stop .
2	C	A question mark is missing: <i>Did you know that pizza is technically a sandwich?</i>
3	A	The word <i>Wall</i> is part of the name of the landmark, so needs a capital letter .
4	N	There are no mistakes. The question mark is needed before the closing speech mark in section A, even though it is not the end of the sentence. The comma is also needed in section B to separate the clauses.
5	B	A comma is needed before the character speaks. If speech begins midway through a sentence, the opening speech mark must always be preceded by a punctuation mark of some form (almost always a comma).

6	C	<i>Your fingerprints are unique: it's like having a secret code on your hands!</i> Fingerprints are not <i>beneath</i> , <i>around</i> or <i>beside</i> hands. They are also not <i>throughout</i> hands (we could maybe say that they are <i>throughout</i> our fingertips, but certainly not our hands).
7	A	<i>He and his friends didn't think that it was a good idea to go the club.</i> A common mistake would be to choose 'Him', but imagine that his friends weren't part of the sentence – what would we write? 'He didn't think...' <u>not</u> 'Him didn't think...'
8	C	Maggie <u>had</u> chosen to remain at home while the others went to the park. The participle must be used because it is preceded by the 'helping verb' <i>had</i> . Compare these: I chose / I have chosen .
9	B	<i>While everyone else was still in bed, Josh had lit the fire, made breakfast and done some shopping.</i> The contrast is between what one set of characters is doing (sleeping) while other characters are doing something different.
10	E	'Unless you change that negative attitude, you'll not be invited to the party at all!' The party has presumably not yet started, so a future tense is needed. You'll is a contraction of <i>you will</i> .

11	B	The spelling should be infinity . Just like the spelling of 'definitely', infinity has 'in it' in it!
12	C	The spelling should be musicians . Other words with the <i>-cian</i> suffix are connected to occupations as well: <i>politician</i> , <i>magician</i> , <i>technician</i> , <i>physician</i> , <i>optician</i> .
13	B	The spelling should be tomorrow . This spelling evolved from an Old English word that literally meant 'to morning' (i.e. the next morning). This may help children remember that there is only one letter m (though they need to remember the double r).
14	C	The spelling should be population . There is no double p. It comes from the same root as the word 'people', which also has no double p.
15	N	There are no spelling mistakes. <i>Business</i> is often misspelled. It may be best to connect it to 'busy-ness' (swapping the y for an i). Job titles, like <i>conductor</i> , are more likely to end with -or (<i>actor</i> , <i>doctor</i> , <i>inspector</i> , <i>director</i> , <i>instructor</i>).

16	D	Line 2 has the answer: <i>When she had gone on several miles she thought she would stop to rest...</i>
17	E	The Scarecrow has a painted face (lines 7-8), shabby clothing (lines 10-11) and was stuffed with straw (lines 7 and 10).
18	D	Although <i>raspy</i> may be too tricky to work with, the word <i>deep</i> is helpful. 'Pointed' and 'deep' are not similar in meaning. Options B and C are adverbs; we are looking for an adjective. 'Deep' makes no sense when swapped with <i>tedious</i> (line 22).
19	A	This passage is about the hope that Dorothy brings to the Scarecrow . The characters meet for the first time, and their initial discussion leads to us learning that he has no brain. However, there is a glimmer of hope that Oz may be able to help.
20	C	Lines 18-50 are an example of direct speech . There are lots of speech marks, and each paragraph focuses on what the two main characters say, with only a little descriptive input from the narrator.
21	B	See line 49 provides the answer: <i>But I do not want people to call me a fool...</i>
22	D	<i>Sadly</i> is an adverb . In line 40 it describes the verb <i>answered</i> .

23	corn	See lines 4-5: <i>...and not far away she saw a Scarecrow, placed high on a pole to keep the birds from the ripe corn.</i>
24	goodbye	See line 1: <i>Dorothy bade her friends goodbye</i> [farewell]... We do not have to know the meaning of <i>bade</i> – we should be able to tell it is a verb, and cannot be replaced with the word <i>farewell</i> .
25	Toto	Other characters are mentioned, such as the Great Oz (e.g. lines 36-45) and the Munchkins (line 9), but these characters are <u>not present</u> in the scene. However, Toto is present (line 17).
26	I shall be greatly obliged to you	See lines 25-26. This is an expression of gratitude, made in advance of the kind act. <i>Thank you</i> on line 21 can't be used, as it is not part of a seven-word phrase.
27	confidentially	This word is on line 46. The other adverbs on lines 39-50 (<i>sadly</i> and <i>awfully</i>) have nothing to do with keeping things secret, so children should choose the third adverb on these lines.
28	adjective	These words describe nouns: <i>ripe</i> describes <i>corn</i> ; <i>faded</i> describes <i>clothes</i> ; <i>friendly</i> describes <i>way</i> ; <i>light</i> describes the <i>Scarecrow</i> . They are therefore adjectives .

Warm-Up 4 Maths Answers

29	A	0.75 means 75 hundredths (the digits end in the hundredths place). 75 is three-quarters of 100.
30	C	An isosceles triangle must have exactly two sides of equal length. Only (8,7) will work. The connected X marks will move four squares in, then three squares up. When connected to (8,7) the newly formed sides will be of equal length.
31	C	Children must read the announcement above the images: <u>the cheapest item will be half-price</u> . This will be the Number Slider. £6.68 (Rabbit Drummer) + £7.76 (Space Gun) + £1.94 (Number Slider at half-price) = £16.38 .
32	D	4.05 litres is the same as 4050 ml . As there are 1000 millilitres in one litre, the number 4.05 must be multiplied by 1000. We must keep the zero digit between the 4 and the 5.
33	E	The area of a rectangle is calculated by multiplying length by width. $14\text{cm} \times 4\text{cm} = \mathbf{56\text{cm}^2}$.
34	A	135 minutes is the same as 2 hours 15 minutes. Subtract two hours from 5.55pm to reach 3.55pm. Subtract 15 minutes to reach 3.40pm. In 24-hour format, this is 15:40 .
35	D	Work backwards from 35, and use inverse (opposite) calculations. <u>Subtract</u> 29 from 35 to get 6. <u>Multiply</u> 6 by 6 to get 36 .
36	A	A cylinder has <u>no</u> vertices . A cube has 8 vertices. A hexagonal prism has 12 vertices. A square-based pyramid has 5 vertices. A triangular prism has 6 vertices.
37	E	7^2 means 7 squared, which means 7 times itself. This is 49. $49 - 3.5 = \mathbf{45.5}$
38	C	Multiply 24 by 9 to find out how many sheets of paper Mr Richardson needs: 216. Subtract this from 500 to find out how many pages are left: 284 .
39	B	We must add together each child's total to find the total number of votes: $15 + 18 + 11 + 12 + 16 = 72$. One-quarter of 72 is <u>18</u> . It was Maeve who had 18 votes.
40	B	$187\text{cm} + 156\text{cm} + 151\text{cm} + 143\text{cm} + 130\text{cm} = 767\text{cm}$. There are 100 centimetres in one metre, so this total is the same as 7.67m .
41	A	Bobby has saved eleven 20p coins so far, which is £2.20. Subtract this from £6.80 to find out how much more he needs to save: £4.60. It would take 23 more coins to reach this target. Instead of $460 \div 20$, children could perform $46 \div 2$.
42	C	All of the angles on a straight line must add to make 180° . There are four 30° angles on the line ($4 \times 30^\circ = 120^\circ$). We need another 60° to reach 180° .
43	C	Look at the arrangements. Pattern 1 has a <u>1</u> x 3 arrangement. Pattern 2 has a <u>2</u> x 4 arrangement. Pattern 3 has a <u>3</u> x 5 arrangement. To continue with this sequence, Pattern 4 must have a <u>4</u> x 6 arrangement: 24 .
44	B	The pencil now measures 13cm and 7mm. This is 3cm and 3mm less than the original length of the pencil. As the image shows, there are 10 millimetres in 1 centimetre, so the pencil is now 33mm shorter than before.
45	B	Count from the 2 litre line to the 3 litre line. There are five steps. One litre (or 1000ml) divided by five makes 200ml, so each interval is worth 200ml. There are 2 litres 400ml in the jug, which is the same as 2.4 litres .
46	D	Shape A is $\frac{3}{4}$ shaded. Shape B is $\frac{2}{4}$ shaded, which is the same as $\frac{1}{2}$. Shape C is $\frac{2}{4}$ shaded, which can't be simplified. Shape D is $\frac{12}{40}$ shaded, which is the same as $\frac{3}{10}$ (divide both values by six) . Shape E is $\frac{3}{15}$ shaded, which is the same as $\frac{1}{5}$.
47	C	A three-quarters anticlockwise turn will bring us to the same compass point as a one-quarter clockwise (or 90°) turn. The answer is northwest .
48	E	Finding the difference usually involves subtraction. However, when we subtract 3777 from 4381 we get 604, which is not an option. Our missing number must be 3777 <u>more than</u> 4381 (the difference would still be 3777). The answer is 8158 .
49	E	One tenth less than 1000 is <u>999 and nine-tenths</u> , which is the same as 999.9 .
50	E	Prisms have two matching shapes that are connected via rectangles (or squares). There are two triangles in the net, and one rectangle to connect each matching side of the triangles. It is a triangular prism .

51	18 jars	One kilogram is 1000 grams, which means 250g will fit into it four times. All we need to do then is multiply 4.5 by 4 to find out how many jars can be filled; it will be 18 of them.
52	60°	Angles next to each other on a parallelogram (or rhombus) will always add together to make 180° . Angle A must therefore be 60° ($180^\circ - 120^\circ$).
53	16:35	95 minutes is 1 hour 35 minutes. Add one hour to 3pm to reach 4pm. Add a further 35 minutes to reach 4.35pm. This is 16:35 in 24-hour format.
54	32 cm	If the area of a square is 64cm^2 then each side must measure 8cm ($8\text{cm} \times 8\text{cm} = 64\text{cm}^2$). Four sides measuring 8cm will make a perimeter of 32cm .
55	0.76	Dial A is pointing directly at 0.2. Dial B is pointing directly at 0.56. $0.2 + 0.56 = \mathbf{0.76}$
56	10 games	Team 1 v Team 2; Team 1 v Team 3; Team 1 v Team 4; Team 1 v Team 5; Team 2 v Team 3; Team 2 v Team 4; Team 2 v Team 5; Team 3 v Team 4; Team 3 v Team 5; Team 4 v Team 5.

Warm-Up 5 English Answers

1	C	The comma needs to be placed <u>before</u> the closing speech mark. In other words, the comma and the closing speech mark are in the wrong order.
2	B	<i>Dreams</i> should not have an apostrophe . It is a simple plural. It is not short for 'dream is' (or 'dream has'). There is no need for a possessive apostrophe either, as the dreams do not own anything.
3	N	There are no mistakes. The two contracted words (<i>couldn't</i> and <i>you'd</i>) both have apostrophes in the correct place. The exclamation mark correctly precedes the closing speech mark.
4	D	There is no full stop at the end of the sentence. It should be placed before the closing speech mark.
5	B	<i>Planet</i> should not be capitalised . It is a common noun, not a proper noun. Only the actual names of the planets get capital letters.

6	E	The police <u>have</u> taken a note of the incident and will deal with the matter as soon as possible. We cannot choose <i>took</i> or <i>wrote</i> because the preceding word is <i>have</i> . In these cases, we would have to say 'have taken' and 'have written'.
7	A	<i>They're not going to be attending the party, even though the whole thing was their idea in the first place.</i> Their is the possessive form of the pronoun 'they'.
8	A	<i>The lost city was hidden among the various dense jungles of South America.</i> A city would have to be the same size as all of the jungles combined if it were to be <i>throughout</i> them! It can't be <i>between</i> them: the word <i>various</i> suggests more than two jungles.
9	C	<i>Except for Orla, everyone had brought their permission slip that morning.</i> Except means 'apart from'. We cannot choose <i>apart</i> or <i>aside</i> because these words must be followed by the word <i>from</i> (not <i>for</i>). <i>Besides</i> cannot be followed by <i>for</i> .
10	B	<i>Let's begin our day by reminding ourselves of what we learned yesterday.</i> Let's needs the apostrophe because it is a contraction of 'let us'.

11	B	The spelling should be disruption . A double p before a <i>-tion</i> suffix is unlikely (if not impossible). Consider examples such as 'option' and 'eruption'.
12	D	The spelling should be knowledge . It has a <i>-dge</i> spelling for its final sound. Children can easily make mistakes with this spelling, particularly as 'college' is not spelled the same way.
13	D	The spelling should be vegetarian . The fourth letter is an e, not an i. Compare it with the spelling of <i>vegetable</i> .
14	C	The spelling should be excitement . The e at the end of 'excite' is <u>not</u> removed. This is because the suffix <i>-ment</i> does not start with a vowel.
15	A	The spelling should be lightning . Although <i>lightening</i> is a word, it means 'making something less heavy or less dark'. It is not the spelling of the natural phenomenon that accompanies thunder.

16	B	Lines 4-20 (the first main section of the report) detail the four basic forces of flight: lift, weight, thrust and drag .
17	D	See lines 9-10: <i>The shape of the wing, <u>called an airfoil</u>, plays a crucial role in lift generation.</i>
18	D	The takeoff process is described in lines 21-33. The increase of the wing's angle is mentioned in lines 30-31, which is part of the Lift Generation section.
19	A	<i>Takeoff</i> is a compound word , which is a combination of two words ('take' and 'off') into one, without the use of hyphens or apostrophes, and without the removal of any letters.
20	C	The two words with opposite meanings are <i>ascend</i> and <i>descend</i> (line 41). <i>Ascend</i> means to rise or go up. <i>Descend</i> means to go down.
21	A	See lines 51-52: <i>Just before touching down, the pilot <u>flares</u> the plane, raising the nose slightly to reduce descent rate.</i>
22	A	In this sentence, <i>brakes</i> is a noun . They are <u>things</u> that the pilot applies pressure to so that the plane comes to a complete stop.

23	acceleration	The spelling of this word is made somewhat easier by the clue: only the first two letters of <i>deceleration</i> are changed. The <i>de-</i> becomes <i>ac-</i> .
24	counteracts	The word is on line 15. Thrust ... counteracts [works against] <i>the drag that tries to slow the plane down</i> . The letter s in <i>works against</i> is a clue that we are looking for a verb that ends with s.
25	crucial role	These words are on line 10. <i>The shape of the wing... plays a crucial role [important job] in lift generation.</i>
26	skiing	Although a double i is hardly ever used in English, it is only because very few verbs end with the letter i. We cannot remove an i from <i>taxiing</i> or skiing as they would sound very different.
27	steady altitude	See line 39. Altitude refers to the plane's height above the Earth. Steady suggests that something is not changing.
28	adjective	These four words describe nouns. <i>Basic</i> describes the noun <i>forces</i> ; <i>greater</i> describes the noun <i>thrust</i> ; <i>next</i> describes the noun <i>step</i> ; <i>complete</i> describes the noun <i>stop</i> .

Warm-Up 5 Maths Answers

29	C	Twenty-five minutes to eight, as a 12-hour time, is 7.35pm. As a 24-hour time, the hour number needs to be increased by twelve from 1pm onward: 19:35 .
30	E	All of these nets can be folded to make cubes. Notice how each has a line of four squares, with one extra square on either side of the line. These formations will always fold to make cubes.
31	E	Each interval along the horizontal axis is worth 15 minutes. Locate the interval after 1pm and scan up from here to the graph line. Look across to the left to find the measurement. It is 26°C , as each interval on the vertical axis is worth 2°C.
32	E	The decimal digits must reach the <u>hundredths</u> place as $\frac{1}{4}$ cannot be converted into tenths. 25 is $\frac{1}{4}$ of 100, so the decimal value is 0.25 .
33	D	Rectangle A's area is 14cm^2 , and its perimeter is 18cm. Rectangle B's area is 15cm^2 , and its perimeter is 16cm. Rectangle C's area is 12cm^2 , and its perimeter is 16cm. Only option D is correct.
34	B	Turn 45° <u>anticlockwise</u> from southwest to discover where Joah began: south .
35	D	Primes are numbers that only have two factors: 1 and something else. 51 is not prime as it can be divided by 3. Any multiple of 3 is easily spotted: its digits add to make a multiple of 3 (e.g. $5+1$ for 51).
36	C	Convert 2.8kg into grams: 2800g. Divide by 8 to discover how much the dog eats in one day: 350g. Now multiply by ten to discover how much it eats in ten days: 3500g .
37	E	30 is 25% (or one-quarter) of Leah's number. Leah's number must be 120 (30×4). 30 is 20% (or one-fifth) of Josh's number. Josh's number must be 150 (30×5). Add these together to get 270 .
38	C	To find the mean, we add all the values then divide the sum by how many values were in the list. $142\text{cm} + 175\text{cm} + 183\text{cm} + 156\text{cm} = 656\text{cm}$. $656\text{cm} \div 4 = 164\text{cm}$. When converted into metres, this value is 1.64m .
39	D	The set of directions will lead us along the furthest west (or furthest 'left') path on the grid. The missing two instructions are North 2 squares & West 3 squares .
40	B	Children should not confuse a 'double-flavour' with two separate types of chocolate. There are 40 chocolates in the box. There are 8 White Chocolate & Nougat Squares. $\frac{8}{40}$ is the same as $\frac{2}{10}$, which is the same as 20% .
41	B	The top two angles in this isosceles are matching. The three angles in a triangle add to make 180° . Subtract 42° from 180° to get 138° . Divide this by two to get the size of one angle at the top: 69° .
42	C	Only option C will create a square, as each side will be five units long.
43	C	10 has four factors: 1 & 10, 2 & 5. 11 has two factors: 1 & 11. 12 has six factors: 1 & 12, 2 & 6, 3 & 4 . 13 has two factors: 1 & 13. 14 has four factors: 1 & 14, 2 & 7.
44	B	$96 \times 7 = 672$ $672 - 98 =$ 574
45	E	None of the angles can be 60° . They must add to make 360° . Option A: $60^\circ + 30^\circ + 15^\circ + 7.5^\circ = 112.5^\circ$. Option B: $120^\circ + 60^\circ + 30^\circ + 15^\circ = 225^\circ$. Option C: $240^\circ + 120^\circ + 60^\circ + 30^\circ = 450^\circ$. Option D: $480^\circ + 240^\circ + 120^\circ + 60^\circ = 900^\circ$. (The angles are actually 192° , 96° , 48° and 24°)
46	C	If we know the length of one side of a square, then the perimeter is this measurement multiplied by four. $8\text{cm} \times 4 =$ 32cm .
47	D	60 is a fraction of another number, which means we are looking for something <u>greater than 60</u> . As it is $\frac{1}{4}$ of a certain number, we must 'divide by the top, multiply by the bottom' to find that number. $60 \div \frac{1}{4} = 15$. $15 \times 5 =$ 75 .
48	D	The first five cube numbers are 1, 8, 27, 64 and 125 ($1 \times 1 \times 1$, $2 \times 2 \times 2$, etc.). The only cube number in the list is 8 .
49	A	Multiply each numerator by four. You should discover that only $\frac{3}{4}$ will not reach (or exceed) its denominator. Every other fraction's numerator would be greater than its denominator. This is the fastest way to discover that $\frac{3}{4}$ is smallest .
50	C	Each pattern is connected to square numbers. The shading helps. Pattern 2 shows 2^2 with 1^2 ($4+1$). Pattern 3 shows 3^2 with 2^2 ($9+4$). Pattern 4 shows 4^2 with 3^2 ($16+9$). Pattern 5 will have to show 5^2 with 4^2 ($25 + 16 =$ 41).

51	£1.85	Subtract the £5.20 change from £20 to find the cost of eight jars: £14.80 . Divide £14.80 by 8 to find the cost of one jar: £1.85 .
52	448 seats	$28 \times 16 =$ 448 . If children struggle with long multiplication, they could double the 28 and halve the 16. The numbers become 56 and 8. 56×8 gives the same answer. Doing this again creates 112×4 .
53	5	1 and 4 are square numbers <u>and</u> factors of 24 so go in the central section. 2 and 3 are factors of 24 so go in the left circle. 5 is neither of these so cannot be placed inside the Venn Diagram.
54	15:00	Add 1 hour then 35 minutes to 9.30am. This takes us to 11.05am. Add two and a half hours to this to reach 1.35pm. Add 1 hour then 25 minutes to this to reach 3.00pm, which is 15:00 .
55	1 line of symmetry	The only line of symmetry is a vertical one going down through the centre. The shape can then be folded across the way.
56	8800	The largest number is 8.8. The number of decimal digits doesn't matter (our question should be 'which one has the greatest value in the tenths place?') $8.8 \times 1000 =$ 8800

Warm-Up 6 English Answers

1	C	Freya has spoken one single sentence, which has been separated by the narrator. Her sentence was as follows: <i>The green ones are his, but those purple ones are definitely hers.</i> In other words, there should not be a capital B on the word <i>but</i> .
2	N	There are no mistakes. The pair of brackets is acceptable (although a pair of commas or dashes would also work). <i>Everyone's</i> needs a possessive apostrophe, and the apostrophe at the end of <i>boys'</i> is also acceptable (plural possession).
3	D	A question mark is needed . Isla asked a question: ' <i>...isn't it?</i> '
4	A	The question mark after <i>question mark</i> is incorrect! Mrs James is making a statement about question marks; she is not asking a question.
5	B	A comma is needed after the word <i>minutes</i> . This stops the words <i>four minutes</i> and <i>stirring</i> from running together, which could create a misunderstanding: should I be stirring for four minutes?

6	A	' <i>You should always be kind to others, no matter what they say or do to you.</i> ' None of the other options work with the words that come after the phrase.
7	D	<i>His understanding of grammar is impressive. Furthermore, he is an excellent reader.</i> The other options are 'contrast words' but the report is stating two positive things, not contrasting a positive with a negative. <i>Furthermore</i> is a formal way of saying 'and'.
8	C	<i>The teacher cancelled the quiz on account of so many students being absent.</i> This phrase has the same meaning as 'as a result of'.
9	B	<i>It was eight o'clock, and Maggie hadn't even begun her Spanish homework.</i> When <i>had</i> (or <i>has</i> , <i>have</i> , etc.) appears before another verb, we must use a past participle. This means that the verb often changes. Compare these: I began / I have begun .
10	E	' <i>I expect each of you to participate in the class discussion.</i> ' Option D would only work if the phrase <i>much of you</i> were changed to <i>many of you</i> .

11	C	The spelling should be research . The prefix is <i>re-</i> , which is attached to the word <i>search</i> .
12	A	The spelling should be excellent . Children often miss the letter c in this word.
13	N	There are no mistakes. The word <i>exhausting</i> contains an h, which is often missed.
14	B	The spelling should be enormous . The root of its spelling is connected to the word <i>normal</i> . Things that are <i>enormous</i> are not of <i>normal</i> size.
15	C	The spelling should be criminals . There is no double m in the word. It comes from the same root as the word <i>crime</i> (which is a clue to there being only one m).

16	E	The poet is trying to create an absurd mood. The information in the poem is clearly ridiculous and not meant to be taken seriously. For example, birds don't exist (line 9), the oysters have shoes (line 23), the sea is boiling hot (line 33).
17	E	See line 3. It was sunny in the middle of the night (another piece of evidence to support the idea that the poem is absurd).
18	B	See lines 11-12. The Walrus and the Carpenter were upset over there being so much sand .
19	E	Lines 22-26 clarify how many oysters went for the walk. On three occasions, we read that four oysters joined in (making twelve), after which we read that <i>more, and more, and more</i> came along. There were more than twelve of them.
20	C	See lines 34-36. The Oysters cry out <i>wait a bit... for some of us are out of breath</i> . The Carpenter replies 'No hurry!', and they thank him for this. This is suggesting that the Oysters wanted to have a rest .
21	C	In lines 46-47, the Walrus is pleading with the Carpenter that they not eat the Oysters. However, in line 50, it is the Walrus who is sorting out the Oysters according to size (which is inferring that he's going along with the plan anyway).
22	D	Line 51 contains the hyphenated word: <i>pocket-handkerchief</i> . The other lines contain a longer 'stroke', which is a dash. Dashes are used to <u>separate</u> phrases whereas hyphens <u>connect</u> two or more words into one.

23	hungrily	Many adverbs add the suffix <i>-ly</i> to the adjective form. However, if the adjective ends with a single y (e.g. <i>sulky, hungry, angry, happy</i>), this is exchanged for an i (<i>sulkily, hungrily, angrily, happily</i>).
24	the eldest Oyster	Lines 19-21: The eldest Oyster looked at him, but never a word he said. The eldest Oyster winked his eye, and shook his heavy head – Meaning to say <u>he did not choose to leave the oyster-bed</u> .
25	shoes	See lines 23-24: <i>...their shoes were clean and neat - and this was odd, because, you know, they hadn't any feet.</i>
26	sealing-wax	Accept <i>wax</i> as an answer, even though the hyphen has connected these two words to make one. To close an envelope is to 'seal' it.
27	sympathise	There are two words in lines 49-51 that rhyme with 'size'. These are sympathise and <i>eyes</i> . However, children have been asked to find the <u>verb</u> . <i>Eyes</i> is a <u>noun</u> on line 51.
28	noun	These are all things, which makes them nouns. Words like <i>chat</i> and <i>treat</i> can often be verbs but, in this poem, they are things: <u>the treat... our chat</u> . Words like 'the' and 'our' precede nouns.

Warm-Up 6 Maths Answers

29	C	50% is the same as $\frac{1}{2}$. In options A, D and E, the numerators are half the denominators, which means these are also equal to $\frac{1}{2}$. However, in option C, 17 is not half of 32, which means $\frac{17}{32}$ is not equal to 50%.
30	E	If 25 cupcakes are to be baked, and we have been given the details for baking 10, then we are multiplying everything by 2.5. However, it is not necessary to be so precise. Just look for the measurement that is <u>between two and three</u> times larger.
31	E	It is a waste of time to perform precise calculations. Round the price of the beach ball to £4, and the price of the bucket and spade to £6. Multiply these by 30 and 20 respectively (rather than 31 and 19). These both make £120, so the answer is £240 .
32	A	8^2 means 8×8 , which is 64. Subtract 8 from this to get 56. This is also the answer to 7×8 .
33	E	Volumes of cuboids are calculated by multiplying each of the three measurements. $3\text{cm} \times 5\text{cm} \times 12\text{cm} = \mathbf{180\text{cm}^3}$
34	B	$\pounds 84.80 + \pounds 47.20 + \pounds 60.05 = \pounds 192.05$ $\pounds 250 - \pounds 192.05 = \mathbf{\pounds 57.95}$
35	B	The scale tells us that the kilometre value is six times greater than the centimetre value. We must therefore make 14.4 six times smaller: $14.4 \div 6 = 2.4$. The answer is 2.4cm .
36	A	Nobody expects children to know how to divide by 139! This is a test of estimating. It would not take many 139s to fit into 417. By adding 139 to itself three times , children will see that this makes 417.
37	A	'Two years from today, Charlie will be nine years old.' This means he is seven years old today. Anna is two years younger than Charlie, so she is five. Casey is seven years older than Anna: Casey is 12 years old .
38	A	Study the first three numbers in the sequence. 61 becomes 58: this is subtract three. 58 becomes 52: this is subtract six. We are either increasing our subtracted amounts by three each time, or doubling the amount. Now test the options.
39	C	12 students out of 48 is equal to one quarter of the total. One quarter of 360° is 90° (a right angle). This corresponds to Charlie and the Chocolate Factory .
40	B	60% is equal to six-tenths (or three-fifths). So, if three-fifths take part in the table quiz, and one-fifth plays volleyball, this leaves us with one more fifth who play video games. 8 teenagers play video games, which is one-fifth of the total: 40 .
41	B	The first five cube numbers are 1, 8, 27, 64 and 125 ($1 \times 1 \times 1$, $2 \times 2 \times 2$, etc.). Only <u>one</u> of the eight cards has a cube number (27). This makes it unlikely (less than half a chance, but not impossible) that a cube number is chosen.
42	E	The area of Rectangle A is 78cm^2 ($13\text{cm} \times 6\text{cm}$). The area of Rectangle B is 45cm^2 ($15\text{cm} \times 3\text{cm}$). The difference between these areas is 33cm^2 .
43	E	Alex gets five points for each correct answer. He answered eight questions correctly, giving him 40 points. However, he also gets one point for each of Joe's incorrect answers. Joe got three questions wrong, bringing Alex's total to 43 .
44	C	The shape has two lines of symmetry : one vertical line down through the centre, and one horizontal line across the centre.
45	E	Perpendicular lines meet at a right angle. The only line that meets Line F at a right angle is Line E .
46	D	The combined weight of the teddy bear and submarine is 400g. The horse weighs 120g, so we need another 280g ($400 - 120$) to balance the scales. The only option that adds to make 280g is two dolls (140g each).
47	E	When numbers are multiplied by 1000, each digit moves three places to the left. The 5 units therefore become 5 thousands, and the 3 hundredths become 3 tens. The answer is 5030 .
48	C	A square-based pyramid has one square face on the base, and four triangles rising to meet at the apex. It therefore has 5 faces in total.
49	B	$\frac{53}{100}$ means 53 hundredths, which is straightforward to write as a decimal number as there is a place known as the hundredths. The digit 3 will be in the hundredths place. The answer is 0.53 .
50	B	A right angle is exactly 90° . If Angle A measures 74° then Angle B must be the difference between 90° and 74° . $90^\circ - 74^\circ = \mathbf{16^\circ}$.

51	30 cm^2	Areas of triangles are calculated by multiplying the base by the height, then dividing by two. Easier still, we can halve the 12cm measurement (6cm) at the beginning, and then multiply by 5.
52	1 minute 43 seconds	Violet took 26 seconds. Heidi took 25 seconds. $24 + 28 + 26 + 25 = 103$ 103 seconds is the same as 1 minute 43 seconds .
53	13	Reading horizontally (to the left of) Sutton, we see the numbers 5, 3 and 1: a total of 9. However, we must also include their match against Euston, which is below the 'vertical' Sutton: 4. $9 + 4 = \mathbf{13}$
54	275	The fifth square number is 25 (5×5). The first five prime numbers are 2, 3, 5, 7 and <u>11</u> . $25 \times 11 = \mathbf{275}$
55	2.505 litres	There are 3500ml in the jug. The three glasses' capacities add to make 995ml ($400 + 375 + 220$). $3500\text{ml} - 995\text{ml} = 2505\text{ml}$. This is the same as 2.505 litres .
56	17	To find the mean, we add the values then divide the sum by how many numbers there were. The total is 102 which, when divided by six, makes 17 .

Warm-Up 7 English Answers

1	C	This is a list of three items. There must be a comma between the first and second items: the <i>chess boards</i> and the <i>Cluedo games</i> .
2	B	The possessive apostrophe is in the wrong place . It should be <i>Jessica's</i> . The apostrophe is always placed directly after the name of the owner or owners. An s is added afterwards if the owner's name did not end with s.
3	D	The exclamation mark should be <u>before</u> the closing speech mark, not after it.
4	A	An apostrophe for contraction is missing. The word should be <i>what's</i> . This is a contraction of <i>what is</i> : ' <i>What is the point in learning about symmetry?</i> '
5	C	The word <i>thousands</i> should not have an apostrophe . It is not a contraction of 'thousand is' (or 'thousand has'), nor does it need a possessive apostrophe as the <i>thousands</i> do not own anything.

6	D	<i>The local bank chose Miss Foster to be their new manager because she had <u>a</u> wealth of experience.</i> The clue is in the preceding word <i>a</i> . We would not say <i>a abundance</i> , nor <i>a lots</i> , nor <i>a years</i> , etc.
7	E	' <i>Your fingerprints are everywhere, Max. We need to destroy the evidence quickly!</i> ' All of the other options would have to be followed by a preposition (e.g. <i>get rid of</i> , <i>get lost in</i>).
8	B	<i>Just as night follows day, it is a certainty that Ella will come home whistling a happy tune.</i> Certainty is the only noun among the options. <i>Possibly</i> and <i>certainly</i> are adverbs. <i>Possible</i> and <i>certain</i> are adjectives.
9	C	<i>Ollie's group <u>had taken</u> too much time to complete the art project.</i> We must use the past participle form of <i>take</i> because it is preceded by the verb <i>had</i> (Ollie's group took / Ollie's group had taken).
10	A	' <i>Please remain seated until you receive further instruction.</i> ' The verb <i>sat</i> refers to the actual moment/action of sitting. But when someone is already in that position – and is expected to remain there – they are seated .

11	D	The spelling should be narrator . Many job titles end with <i>-or</i> rather than <i>-er</i> (e.g. doctor, professor, navigator). It is not a foolproof spelling rule by any means though (e.g. farmer, carpenter, writer).
12	C	The spelling should be designers . If we look at the misspelled word too quickly, we might fail to notice that the letters in the middle spell 'sing' rather than 'sign'.
13	A	The spelling should be reliable . Words that end with a single y (<i>rely</i>), and no preceding vowel, will exchange the y for an i when a suffix is added.
14	N	There are no spelling mistakes. <i>Strength</i> is often misspelled, but the <i>-ng</i> and <i>-th</i> sounds can clearly be heard. <i>Despair</i> is often mistakenly spelled as 'dispair' or 'dispare'.
15	B	The spelling should be extremely . In order to help children spell <i>extreme</i> correctly, point out that the letter e appears at either <i>extreme</i> .

16	B	The character is not introduced until the second paragraph. The first paragraph is describing the world in which the main character lives (or, in this case, had lived).
17	C	Bambi is inquisitive , which means he is constantly asking questions (we can hear the word 'quiz' when we say the word). Line 12 alone is enough evidence for this: <i>Bambi questioned her. He loved to ask his mother questions.</i>
18	D	Bambi can see <i>the whole heaven</i> (or sky). There is no shade of trees. Line 38 later informs us that these things are taking place at the meadow's edge .
19	E	It is <i>early summer</i> (line 1) at the beginning of the passage. But we can tell that some time has passed by the end of the passage as the leaves are now falling from the trees (lines 38-44).
20	A	The crow's words (lines 33-35) suggest a change in tone. A prince had died (although, within this passage, we can't be sure who this is). The desperate clinging of the leaves to the tree (lines 40-42) are also a contrast to the positive opening (lines 1-11).
21	C	Intentionally means to do something with intention, or on purpose.
22	E	The positioning of the possessive apostrophe at the end of <i>thrushes'</i> and <i>yellowbirds'</i> tells us that there are multiple of each. By contrast, the positioning of the apostrophe before the end of <i>owl's</i> , tells us that there is only one owl.

23	5	The five colours are <i>blue</i> (line 1), <i>red</i> , <i>white</i> , <i>yellow</i> (line 3) and <i>green</i> (line 8).
24	fragrant silence	<i>Scented</i> is connected to fragrant . <i>Hush</i> is connected to silence . The phrase is on line 11.
25	boundless	This word is on line 28. It means 'without boundary', 'limitless' or 'unending'.
26	length	Adjectives such as <i>warm</i> , <i>long</i> , <i>strong</i> , <i>wide</i> , <i>deep</i> and <i>broad</i> will use a <i>-th</i> suffix when converted into nouns. Sometimes the vowels change too: <i>warmth</i> , length , <i>strength</i> , <i>width</i> , <i>breadth</i> , <i>depth</i> .
27	It isn't the way it used to be	This phrase appears on line 42 and is 'spoken' by one leaf to another.
28	verb	Each of these words is a past tense verb .

Warm-Up 7 Maths Answers

29	D	There are 15 days left in October, then another two to reach 2nd November. This makes 17 days altogether, which is two weeks and <u>three days</u> . All we need to do then is count three days after Monday. We will reach Thursday .
30	A	This is a kite . Its two longer sides meet, as well as its two shorter sides. The two longer and shorter sides must also have matching measurements. There are no parallel lines, and there is one line of symmetry (tilt your head to the right!).
31	D	Convert both measurements into centimetres. They become 225cm and 110cm (there are 100 centimetres in one metre). The difference is 115cm .
32	C	This isn't as tricky as it seems because each option can be tested out. The fastest way, however, is to add 50p and 20p to make 70p, then work out how many times this fits into £5.60. It fits eight times, which means eight of each coin: 16 altogether.
33	C	The six scores need to be added, then the total is divided by six. $8 + 6 + 8 + 7 + 9 + 7 = 45$ $45 \div 6 = 7.5$
34	E	The four angles of a quadrilateral must always add to make 360° . The three given angles only add to make 144° , which leaves us with another 216° to account for. This huge angle must therefore be a reflex angle (between 180° and 360°).
35	D	The number of decimal digits is not important. Which number has the largest digit in the tenths place? 0.2
36	C	The difference between the cost of each stamp is 35p. The difference between the cost of eight of each can be found by multiplying 35p by eight: £2.80 .
37	D	There are 28 girls ($70 - 42$). $\frac{28}{70}$ can be simplified to become $\frac{2}{5}$. This is the same as $\frac{4}{10}$, or 40% .
38	B	Bottles of soft drink can come in handy as a reference point – two-litre bottles for example. Children should be able to recognise that they can set much more than five of these bottles in a bath. This makes 100 litres the likeliest answer.
39	A	Each horizontal row of numbers adds to make 34. According to the rules, each vertical column has to do the same. The order, from left to right, is 9, 7, 6, 12 .
40	C	Although two of the options (36 and 100) are both even numbers and square numbers, only one of them is part of the <u>first five</u> multiples of nine: 36 .
41	E	All of the numbers are factors of 48. There are ten factors of 48: 1 & 48, 2 & 24, 3 & 16, 4 & 12, 6 & 8.
42	B	Only Nets B and C can be folded to make cubes. They each have a line of four squares, with the two others positioned on either side of the line.
43	E	A quarter to eight in the evening is 7.45pm, which is 19:45. 75 minutes is 1 hour and 15 minutes. This brings us back to 6.30pm, which is 18:30 .
44	C	Imagine the shape as a full rectangle, with its top-right corner 'restored'. The area would be 44cm^2 (11×4). The two measurements of the removed corner are a 2cm width and a 3cm length ($2\text{cm} \times 3\text{cm} = 6\text{cm}^2$). $44\text{cm}^2 - 6\text{cm}^2 = \mathbf{38\text{cm}^2}$
45	D	Michael cannot have three numbers making a sum of 299 because each number is different. His highest possible sum would be 297 ($100 + 99 + 98$).
46	D	The number of lines used in each pattern is <u>two more</u> than the previous pattern (not three). The sequence is 3, 5, 7, 9, 11, 13, 15 . Perhaps some children will spot that the 'pattern number doubled, then plus one' will also give you the number of lines.
47	C	Focus on the simplest fraction: $\frac{3}{4}$. Which other options are larger? 23 is less than $\frac{3}{4}$ of 32, so $\frac{23}{32}$ is smaller. 5 is less than $\frac{3}{4}$ of 8, so $\frac{5}{8}$ is smaller. 13 is less than $\frac{3}{4}$ of 20, so $\frac{13}{20}$ is smaller. 13 is <u>more</u> than $\frac{3}{4}$ of 16, so $\frac{13}{16}$ is the largest fraction in the list.
48	E	The scale of the map tells us that the number of kilometres in real life will be four times greater than the number of centimetres on the map. Divide 32 by four to get the number of centimetres (not kilometres!): 8cm .
49	A	82 tenths, if it were written as a fraction, would be $\frac{82}{100}$. This is the same as $\frac{8}{10}$, which is the same as 8.2 .
50	D	Encourage children to <u>not</u> carry out the actual calculation. Estimations are simpler. Round both numbers to their nearest ten. The calculation becomes 50×50 , which gives us an estimated answer of 2500 (5×5 with two zeros).

51	6.25 km	If Carmel has travelled $\frac{3}{4}$ of the journey, then she has $\frac{1}{4}$ left to travel. Convert to metres if it makes the calculation easier: $10,000\text{m} \div 4 = 2500\text{m}$. $2500\text{m} \times 5 = 12500\text{m}$, which is 12.5km .
52	315°	Calculate the size of the isosceles angle at the centre of the octagon. <u>Both</u> other angles are 67.5° so it must be 45° ($180^\circ - (67.5^\circ \times 2)$). There are 360° in a full turn/circle. Subtract the 45° from this to get 315° .
53	5 m	There may be ten boys but... there are only <u>nine</u> spaces between them! The measurement of 45m must be divided by nine (not ten). The space between one boy and the next is 5m .
54	£60	25% is the same as $\frac{1}{4}$. One-quarter of £80 is £20. This amount is reduced from the original price to create a sale price of £60 .
55	36 cubes	There are four identical layers from front to back. The front layer shows nine cubes. Multiply this by four to get the total number of cubes: 36 .
56	13 minutes	Henry arrives at the platform at 0851. The train information tells us that they depart every 15 minutes. The next one should be at 0900 but is delayed until 0904. Henry must wait 13 minutes .

Warm-Up 8 English Answers

1	B	There should not be a comma between the words <i>much</i> and <i>higher</i> . These words not only belong to the same phrase, but one is describing the other. In these cases, they should not be separated by commas.
2	D	A question mark is needed: '... <i>why is the traffic so heavy?</i> '
3	C	An apostrophe for contraction is needed: ' <i>It's been ten years...</i> ' (or ' <i>It has been ten years...</i> ').
4	A	There should not been an apostrophe in the first word of the sentence: <i>Raindrops</i> . They do not possess anything in the sentence, nor is it a contraction of two words.
5	D	The speech has not been closed. A final speech mark is needed after the full stop.

6	A	<i>By next month, we will have moved into our new house.</i> The phrase needs to include a sense of the future (<i>will</i>) because the action hasn't been completed yet. But it also needs to include a sense of the past, as the action will have been completed <i>by then</i> .
7	C	<i>A letter was sent from here to Kenya one month ago but, as yet, we have received no reply.</i> Children must focus on the words <i>...here to Kenya</i> . <i>Sent off here to Kenya</i> is not a complete phrase. Options D and E mean the same thing, and make no sense.
8	E	<i>There were at least two hundred guests at the wedding.</i> For options A and C to be correct they would have to say <i>more than</i> and <i>fewer than</i> respectively. <i>Greater than</i> and <i>lower than</i> are not grammatically correct when actual numbers can be counted.
9	D	<i>Whether you agree with the decision or not, it has already been made.</i> Swap the two clauses around to see why only option D works: <i>The decision has already been made, whether you agree with it or not.</i>
10	B	<i>Ruth brought her dog along with her to the school playground. Everyone was delighted.</i> Brought is the past tense of <i>bring</i> . There are no such words as <i>brang</i> or <i>brung</i> . <i>Bought</i> is the past tense of 'buy'.

11	B	The spelling should be quarters . It may sound like it should have an o, but an a is used instead.
12	D	The spelling should be description . The second letter is part of a syllable that is unstressed (spoken quickly), which makes it impossible to determine the letter when we hear the word.
13	N	There are no spelling mistakes. <i>Beginning</i> has a double n due to it being part of the stressed syllable. Note the difference between how <i>opening</i> has no double n (the n is not part of a stressed syllable) but <i>beginning</i> does.
14	C	The spelling should be receive . The <i>i before e except after c</i> rule works for this word. The ie/ei combination of letters make the long ee sound. This means i precedes e, <u>unless the letter c comes directly before them</u> .
15	A	The spelling should be measurement . The final e from the word <i>measure</i> is not removed, as the prefix <i>-ment</i> does not begin with a vowel.

16	D	This passage gives an overview of the success of the Wallace & Gromit franchise. It does not explain how stop-motion works; nor does it give a story of Nick Park's career, nor the back-story for Wallace's character.
17	A	<i>The whimsical world of Wallace and Gromit</i> is an example of alliteration : a purposeful repetition of the same starting sound in words, so as to sound catchy.
18	B	See lines 10-11: <i>...Wallace and Gromit returned to television screens in 'The Wrong Trousers' in 1993.</i>
19	E	The answer is on line 9: <i>'the film was an instant success.</i> It was not a feature-length film; it was a <i>short film</i> (line 6). The character of Feathers McGraw was in a different story (line 12). We only learn that <i>Wallace</i> loves cheese (line 7), not Gromit.
20	B	Options D and E do not sound right when swapped for the word <i>development</i> . Options A to C are tougher. A <i>milestone</i> refers to a single time or achievement, not a development. Hopefully children know the word <i>impact</i> . This leaves us with evolution .
21	C	The passage does <u>not</u> present each show/film in chronological order. Look at lines 17 and 28. 'Cracking Contraptions' aired three years before the 'Were-Rabbit' film.
22	A	Personalities is a noun . Although abstract and invisible, they are still <u>things</u> .

23	canine	See line 3: <i>...his faithful canine companion...</i>
24	international acclaim	The answer is on line 15: international corresponds to <i>worldwide</i> ; acclaim corresponds to <i>praise</i> .
25	lexicon	See line 33. The words lexicon and <i>vocabulary</i> can often be used interchangeably. One word that may catch out some children is <i>catchphrases</i> , but this is a plural (and <i>vocabulary</i> is singular).
26	Were-Rabbit	A hyphen is a punctuation mark that combines two or more words into one. It is a short stroke (similar to the subtraction symbol).
27	1	The only adjective that describes <i>charm</i> is <i>enduring</i> (line 39). <i>Creativity</i> and <i>innovation</i> are both nouns, and are distinct from <i>charm</i> . They are not describing the <i>charm</i> .
28	verb	These are all past tense verbs ('doing' words).

Warm-Up 8 Maths Answers

29	E	$\frac{12}{30}$ is not equal to $\frac{3}{5}$. The numerator (12) is four times greater than the other numerator (3), but the denominator (30) is six times greater than the other denominator (5). This does not balance, meaning the fractions are not equal.
30	A	The number of petals in each pattern increases by three (3 then 6 then 9...). There are four patterns from Pattern 6 to 10, so we just need to multiply by 4 by 3 to work out how many more petals will be in the tenth pattern: $4 \times 3 = 12$
31	E	There are 360° in one full turn. $360^\circ \times 3 = 1080^\circ$. We must then add on the number of degrees in a quarter-turn: 90° . $1080^\circ + 90^\circ = 1170^\circ$.
32	E	Three farms have 1-10 sheep (6, 7, 8). Five farms have 11-20 sheep (12, 15, 15, 18, 19). Five farms have 21-30 sheep (22, 27, 27, 29, 30). Five farms have 31-40 sheep (31, 35, 36, 38, 40). Six farms have 41-50 sheep (41, 46, 46, 47, 48, 50).
33	C	We need to find three measurements to multiply together to make the volume. We have the depth and the height, but not the length (horizontal measurement). As the front face is a square, this measurement must also be 4cm: $4 \times 4 \times 10 = 160$
34	D	8 tens = <u>80</u> . 3 ten-thousands = <u>30000</u> . 8 tenths = <u>0.8</u> . $80 + 30000 + 0.8 = 30080.8$
35	B	Both divisions have the same starting number: 448. The difference is that we are being asked to divide by double the amount this time (16 instead of 8). This means we are <u>halving</u> the first answer: 56 becomes 28 .
36	D	A microwave oven will be considerably heavier than a bag of potatoes (a handy reference point for estimating weight), which typically weighs a few kilograms, but it will still be liftable. 20kg is the most sensible estimate.
37	D	The eighth square number is <u>64</u> (8×8). The first three prime numbers are 2, 3 and <u>5</u> . $64 \times 5 = 320$
38	A	Obtuse angles are greater than 90° but less than 180° . So, if we were to add together two of the smallest obtuse angles ($91^\circ + 91^\circ$), we would still go beyond 180° . Statement A cannot be true.
39	C	There are two faces for the front and back, then count the number of sides that the front face has (six) as this will tell us the number of faces around the side of the shape. There must be eight faces in total.
40	B	$1\frac{1}{4}$ is the same as 1.25. We are therefore looking for a number that is greater than 1.25 but less than 1.5. The only option is 1.35 .
41	C	A kite will have two connecting shorter sides (of equal length) and two connecting longer sides (of equal length). All four sides cannot be the same length, which is why (5,10) makes a rhombus, and not a kite. The answer is (5,7) .
42	C	200 minutes is 3 hours 20 minutes. Although midnight is 00:00 on a 24-hour clock, it's helpful in this calculation to think of it (unofficially!) as 24:00. Subtract 3 hours to reach 21:00, then 20 minutes to reach 20:40 .
43	A	70% of 20 is the same as 7×2 . Look for the name(s) that scored 14. Each interval across the chart is worth two marks. Both Frankie & Dylan scored 14 out of 20.
44	C	April = 30 days; May = 31 days; June = 30 days $30 + 31 + 30 = 91$
45	E	Choosing an odd number (A) is impossible. It is unlikely that a multiple of ten (C) will be chosen as there are only two of them (30 & 60). Half the numbers on the list will be multiples of 12 (B). Most will be less than 40 (D), and all will be even (E).
46	D	The shaded rectangle clearly has a longer measurement of 10cm. As the entire shape is a square, we know that the unlabelled side is also 10cm, one-fifth of which is 2cm. So each rectangle measures 10cm by 2cm. The perimeter is 24cm ($10+10+2+2$).
47	E	One litre is the same as 1000 millilitres. To find $\frac{3}{8}$ of this, we divide by eight (125ml) then multiply by three: 375ml .
48	D	A halfway number is the same as the mean (average) of two numbers. $84 + 120 = 204$. $204 \div 2 = 102$
49	E	0.5kg is the same as 500g (there are 1000g in one kilogram). There are forty £1 coins, each weighing 9g. $9g \times 40 = 360g$. Add this to the 500g to get 860g .
50	C	Although not often taught this way, by definition a prime number is a number with <u>exactly two factors</u> (1 and something else). 17 is a prime number (its only factors are 1 and 17).

51	29	Use inverse (opposite calculations), working from right to left. $52 + 35 = 87$ $87 \div 3 = 29$
52	28°C	The warmest temperature is 21°C . The coldest is -7°C . Although children may know that a range is found by subtraction, this one looks weird: $21 - -7$. This is the same as 21 + 7 .
53	42°	The sum of the angles in <u>any</u> four-sided shape is 360° . There are two 90° right angles, so the other two must add to make 180° . $180^\circ - 138^\circ = 42^\circ$
54	7319	The final digit must be 9 (an odd digit that is a multiple of 3, <u>and</u> must be the largest of the digits). The first digit is therefore 7 (two less than 9). 1 is the only odd non-prime digit left, so comes third.
55	F	Southeast <u>from</u> point A means going diagonally through each bottom-right corner of the squares. We will go through point F.
56	1.4 kg	$175g \times 8 = 1400g$ $1400g = 1.4kg$ (there are 1000 grams in one kilogram)

Warm-Up 9 English Answers

1	B	The word <i>its</i> should not have a possessive apostrophe . We are talking about the mouth that belonged to <i>it</i> , which is a pronoun. Possessive pronouns do not get apostrophes (e.g. my, your, his, her).
2	B	There must be a comma between <i>shoulder</i> and <i>Jake</i> . There is a pause. The word <i>shoulder</i> is the end of the sentence's first (dependent) clause, and the word <i>Jake</i> is the beginning of the second (independent) clause.
3	D	The question mark and the closing speech mark need to switch places. After all, it is Louie's question, not the narrator's. There should always be a punctuation mark <u>before</u> a closing speech mark.
4	N	There are no mistakes. It is one clause, so there is no need for a comma. The possessive apostrophes are correct. Even though <i>cheesecake</i> is not mentioned after <i>George's</i> name, it has appeared earlier in the sentence (we know what the author means).
5	C	<i>December</i> is a proper noun and must be capitalised .

6	E	<i>Although Lauren gets nervous before every performance, she always rises to the occasion once she's on the stage.</i> The statement is not set in the past; it is a present tense sentence. Only use <i>raises</i> when raising something else, not yourself.
7	A	'If I'd known that you were coming, I'd have prepared something to eat.' The verb follows on from the word <i>had</i> (I'd is a contraction of I <u>had</u>). In these cases, we must use the past participle. Compare these: I knew / I <u>had known</u> .
8	C	<i>She couldn't imagine life beyond the small town where she grew up.</i> The sentence tells us she grew up <i>in</i> the town (so she doesn't have to imagine that!). The idea is that she can't imagine life somewhere else, so beyond suits this purpose.
9	D	<i>Our team and theirs were the best in the league.</i> Never use possessive apostrophes with pronouns. It cannot be <i>Us</i> and <i>them</i> . We wouldn't say <i>Us are the best in the league</i> or <i>Them are the best in the league</i> .
10	B	<i>The Gibsons hadn't spoken to the Wilsons for two and a half years.</i> <i>Since</i> could only be used if there were a more specific date mentioned (e.g. 'since last week' or 'since Tuesday').

11	A	The spelling should be sincerely . The final e from <i>sincere</i> is not dropped because the suffix <i>-ly</i> does not begin with a vowel.
12	B	The spelling should be typically . It comes from the word <i>type</i> .
13	D	The spelling should be weird . Some claim that this breaks the <i>i before e except after c</i> rule, but this is not true. This spelling rule almost always works, but only when the ie/ei combination makes a long ee sound. In the word weird , the sound is 'ear' (not ee).
14	A	The spelling should be several . Did any child spot it at the beginning of question 42?
15	N	There are no spelling mistakes. <i>Completely</i> keeps the final e from <i>complete</i> because the <i>-ly</i> suffix doesn't begin with a vowel. The <i>para-</i> prefix in <i>paragraph</i> is common (e.g. <i>parallel</i> , <i>parachute</i>). Mistakes are often made with the second a in <i>character</i> .

16	D	The rhyming pattern of this poem is the basic AABB pattern in which line 1 rhymes with line 2, line 3 with line 4, etc.
17	E	The journey begins in Canada (line 4). Their journey takes them through (or <i>via</i>) Greenland (line 23). Their destination is Strangford (line 29).
18	B	Gaggle is the collective noun for 'geese'. <i>Canopy</i> doesn't work as the geese are <i>beneath</i> it (lines 1-2). <i>Migratory</i> is connected in meaning to 'migrate'. It can't be <i>spectrum</i> (line 10) as the geese don't confront themselves!
19	C	Options A and B are only placed there because <i>boreal</i> sounds like a combination of 'boring' and 'real'. It isn't! Options D and E make no sense when describing a forest. This leaves us with the correct option: Arctic .
20	A	The correct option is once per year . See line 2: <i>Brent geese commence their <u>annual</u> flight</i> .
21	D	The danger in Greenland is hunters (lines 21-28).
22	E	A metaphor compares one thing to another because of a connection. Unlike similes, the words 'like' or 'as' are not used. Line 37 says <i>nature's ballet</i> . The geese are not actually performing ballet but we know what the author means: they are graceful.

23	Mother Nature	See line 12. A two-word proper noun means one name made up of two words, <u>both with capital letters</u> .
24	inventory	The word is on line 32. This question is testing a child's patience in scrolling through the tricky words in order to find one that could possibly refer to the geese being countable.
25	pianos	Unfortunately, there is very little consistency to plurals that end with <i>-oes</i> (e.g. potatoes) and those that end with <i>-os</i> (e.g. logos). These spellings just need to be revised occasionally.
26	marshland	See line 35. A compound word is one word that has been purposefully created by two words joined together, without the use of a hyphen or apostrophe for contraction. <i>marsh</i> + <i>land</i> = <i>marshland</i>
27	goslings	The word is on line 40. For a word that means <i>young geese</i> we are looking for a plural, which means the chances are high that our word ends with <i>s</i> . This narrows our options.
28	noun	In these questions, children should use the word with which they're most familiar (probably <i>day</i>). The other words will be the same part of speech. A <i>day</i> is a <u>thing</u> . It is therefore a noun .

Warm-Up 9 Maths Answers

29	D	If eight meals were served on Wednesday, and only two plates are shown in the pictogram, then a picture of a plate equals four plates. $4\frac{1}{2}$ plates are shown for Friday. $4\frac{1}{2} \times 4 = 18$
30	B	The two triangles comprise <u>half</u> of the rectangle. The area of the rectangle is 48cm^2 . This makes the combined area of the two triangles 24cm^2 .
31	C	Work backwards through the story. Beatrice's number is 432. Barbara's is $\frac{2}{3}$ of this. Divide by three then multiply by two to get 288. Yasmina's number is 88 more than this. $288 + 88 = 376$
32	C	We cannot choose the tenth square number ($10 \times 10 = 100$) as this is not less than 100. There must therefore be nine .
33	D	A fast 'trick' for finding the perimeters of these shapes is to locate the longest horizontal side (11cm) and the longest vertical side (6cm). The perimeter of the shape will be exactly the same as a rectangle with 11cm and 6cm sides: 34cm ($11+11+6+6$).
34	C	In 84 <u>6</u> 50, the 6 digit is in the hundreds place. When dividing by 1000, each digit moves three places to the right. The digit 6 will move to the position immediately after the decimal point: the tenths place: 84. <u>6</u> 5
35	D	Harvey sprints ten times, each sprint lasting 10 seconds. This is 100 seconds, which is 1 minutes 40 seconds. However, he will only need to have <u>nine</u> one-minute rests (we stop counting after the tenth sprint is over). This makes 10 minutes 40 seconds .
36	B	We would have to buy five of the 200g yogurts in order to have one kilogram (1000g). $5 \times 80\text{p} = £4$. The difference in price between this and a one-kilogram container (£3.50) is therefore 50p .
37	A	$8 \times 6 = 48$ $48 - 21 = 27$
38	C	Most people's heights are between 1 and 2 metres. A two-storey house is therefore higher than 1m. Children should look up at the ceiling and ask 'How many of me would it take to touch the ceiling, then again, and then the roof?' The best estimate is 10m .
39	E	2, 6, 8, 10 and 30 are all multiples of 2 . 3, 6, 9, 15 and 30 are all multiples of 3 . 5, 10, 15, 25 and 30 are all multiples of 5 .
40	D	Round 302 to the nearest hundred: <u>300</u> . Round 19 to the nearest ten: <u>20</u> . $300 \times 20 = 6000$ (3 x 2 with three zeros added).
41	C	Just as with parallelograms, angles next to each other on a rhombus add to make 180° . $180^\circ - 76^\circ = 104^\circ$
42	D	If there are eight slices per pizza, then five pizzas would have 40 slices. An extra quarter of a pizza means an extra two slices (two is one-quarter of eight). 42 slices have been eaten altogether.
43	B	Because both the percentage value (40%) and the number (50) end with a zero, we can use the fast 'trick' of ignoring the zeros and simply multiplying 4 by 5. The answer is 20 .
44	C	There are 12 edges on a cube, each of equal length. Divide 300 centimetres (the equivalent of 3 metres) by 12 to calculate the length of one piece of straw: 25cm .
45	B	A 360° turn will bring the hand back to 00. This leaves us with <u>240</u> ($600^\circ - 360^\circ$). There are 30° between each number ($360^\circ \div 12$). How many 30s make 240? Eight. Move clockwise to the eighth number: 40 .
46	A	A triangular prism has 5 faces. A cone has 1 edge. A square-based pyramid has 5 vertices. $5 + 1 + 5 = 11$
47	C	The first four multiples of 4 are 4, 8, 12, and 16. To find the mean of these numbers, add them together (40) then divide by how many numbers were in the list (4). $40 \div 4 = 10$
48	E	$£138.00 \div 4 = £34.50$. If children are using the traditional 'bus stop' method for division, keep an eye on how they use remainders once they reach the decimal places. All too often they assume the final remainder is the same as a decimal digit.
49	D	<i>Twenty minutes to seven in the evening</i> is the same as 6.40pm, which is the same as <u>18:40</u> . The next available screening is at 19:10 (<i>Worst Party Ever</i>), which lasts for <u>90 minutes</u> (1 hour 30 minutes). This movie therefore finishes at 20:40 .
50	E	$1\frac{1}{2}\%$ can be simplified to make $\frac{3}{4}\%$ (divide both values by two). As there is a decimal place for the tenths, things become more straightforward: $\frac{3}{4}\%$ equals 0.7 .

51	2	Rionach counts in 6s, Marco in 8s. The first number they will <u>both</u> say is 24. The next number will be double this: 48. This means their lists will only share two numbers , as they stop at 50.
52	650 ml	It is important to know that $\frac{1}{4}$ of a litre is 250ml ($1000 \div 4$). Count the intervals from one label to the next: there are five. Each interval is worth 50ml ($250\text{ml} \div 5$). We are 100ml below the 750ml mark.
53	isosceles (triangle)	An isosceles triangle has no parallel sides but one line of symmetry. The square goes in box 1, the rhombus in box 2, the isosceles in box 3, and the scalene in box 4.
54	northwest	One quarter-turn (90°) anticlockwise from south brings us to east. A 270° clockwise turn from east brings us to north. A 45° anticlockwise turn from north brings us to northwest .
55	Day 7	Only look at days in which the line has dropped. Each interval on the vertical axis is worth two. If we are looking for a drop of 12 this means we are looking for a drop of six intervals (or lines).
56	13.9 & 20.7	Subtract 3.7 from 7.1. Then subtract 7.1 from 10.5. In both cases, the answer is <u>3.4</u> . The pattern is 'add 3.4'. $10.5 + 3.4 = 13.9$. $17.3 + 3.4 = 20.7$.

Warm-Up 10 English Answers

1	C	A comma is needed after the word <i>students</i> . If direct speech begins in the middle of a sentence, the opening speech mark must be preceded by a comma.
2	A	<i>One's</i> should have an apostrophe for contraction . It is a contraction of the words 'one is': ' <i>This <u>one is</u> mine...</i> '
3	N	There are no mistakes. <i>He'd</i> is a contraction of <i>he would</i> . No comma is needed: the phrase <i>in order to</i> creates a natural, smooth flow.
4	A	Easter is a proper noun and needs a capital letter. The possessive apostrophe at the end of <i>grandparents'</i> indicates that there is more than one grandparent, which is perfectly reasonable.
5	D	Possessive pronouns like <i>his</i> (as well as similar ones like <i>her, hers, yours, ours, their, theirs</i> , etc.) should never have a possessive apostrophe .

6	A	<i>Dad <u>had become</u> tired of Tom's behaviour so decided to do something about it</i> . The helping verb <i>had</i> tells us that we must use a past participle. Although the past tense of <i>become</i> is <i>became</i> , the past participle is <i>become</i> (I become, I became, I had become).
7	C	However <i>much you try to persuade me, my decision won't change</i> . There is a connection between the commonly used phrase 'how much' and the phrase 'however much'.
8	B	Upon <i>the hill stood an ancient castle, overlooking the valley below</i> . If it is overlooking the valley, the castle must be near, or at, the top of the hill. <i>Above</i> the hill might suggest that the castle is floating in the air!
9	A	While <i>the children played noisily in the garden, their grandfather quietly read a book on the porch</i> . The conjunction <i>while</i> is appropriate when describing two events that are happening simultaneously.
10	E	<i>She has few friends who truly understand her</i> . The word <i>much</i> cannot be used here: the opposite of <i>few</i> is <i>many</i> .

11	C	The spelling should be hedgehogs . The final e is not dropped from <i>hedge</i> in this compound word, as <i>hogs</i> does not begin with a vowel.
12	C	The spelling should be straightforward . The word <i>straight</i> contains the <i>-gh</i> letter combination.
13	B	The spelling should be weather , which is a homophone of <i>whether</i> . Compare the two words in this sentence: The forecast was for bad <i>weather</i> so we had to decide <i>whether</i> or not to cancel our plans.
14	N	There are no spelling mistakes. <i>Behaviour</i> has the common <i>-our</i> ending (e.g. colour, honour) but also needs the letter i to create the long ee sound: 'be-hav-ee-our'. When said quickly, this sounds like 'behaiv-yer'.
15	A	The spelling should be busily . It comes from the adjective <i>busy</i> . The <i>-ly</i> suffix is added to create the adverb (<i>busy</i> + <i>ly</i>). However, the letter y at the end of <i>busy</i> must then be exchanged for the letter i.

16	A	Lines 5-9 define what the narrator meant by <i>silent sympathy</i> : it is to show love to someone who is going through a hard time , but to also to know to not talk about their problems as this could be intrusive.
17	D	Lines 21-22 give the reason why Mother wrote the letter: <i>...none of them could tell where the wife and children of Mr Szczepansky would be likely to be</i> .
18	B	Look at lines 14-15: <i>Bobbie always had these thoughts at the back of <u>her</u> mind...</i> Bobbie is speaking to herself . If we read the first nine lines carefully, we will discover that Bobbie is another name for <i>Roberta</i> .
19	C	See lines 26-27: It just means to trying to help people .
20	E	There is no evidence in these lines that they learned Russian, nor that <u>they</u> taught him English. We do not learn whose idea it was to give him Father's clothes, nor that they searched <i>nearby streets</i> . But line 31 tells us that they brought him flowers .
21	A	Line 37 tells us that [Mother] <i>likes to have someone to be good to</i> but that she also <i>hated</i> to let the man have Father's clothes. The good deed was hurting her personally, but she knew it was the right thing to do .
22	A	The phrase on line 5 is <i>rare accomplishment</i> . <i>Rare</i> corresponds to <i>unique</i> . <i>Accomplishment</i> corresponds to <i>achievement</i> .

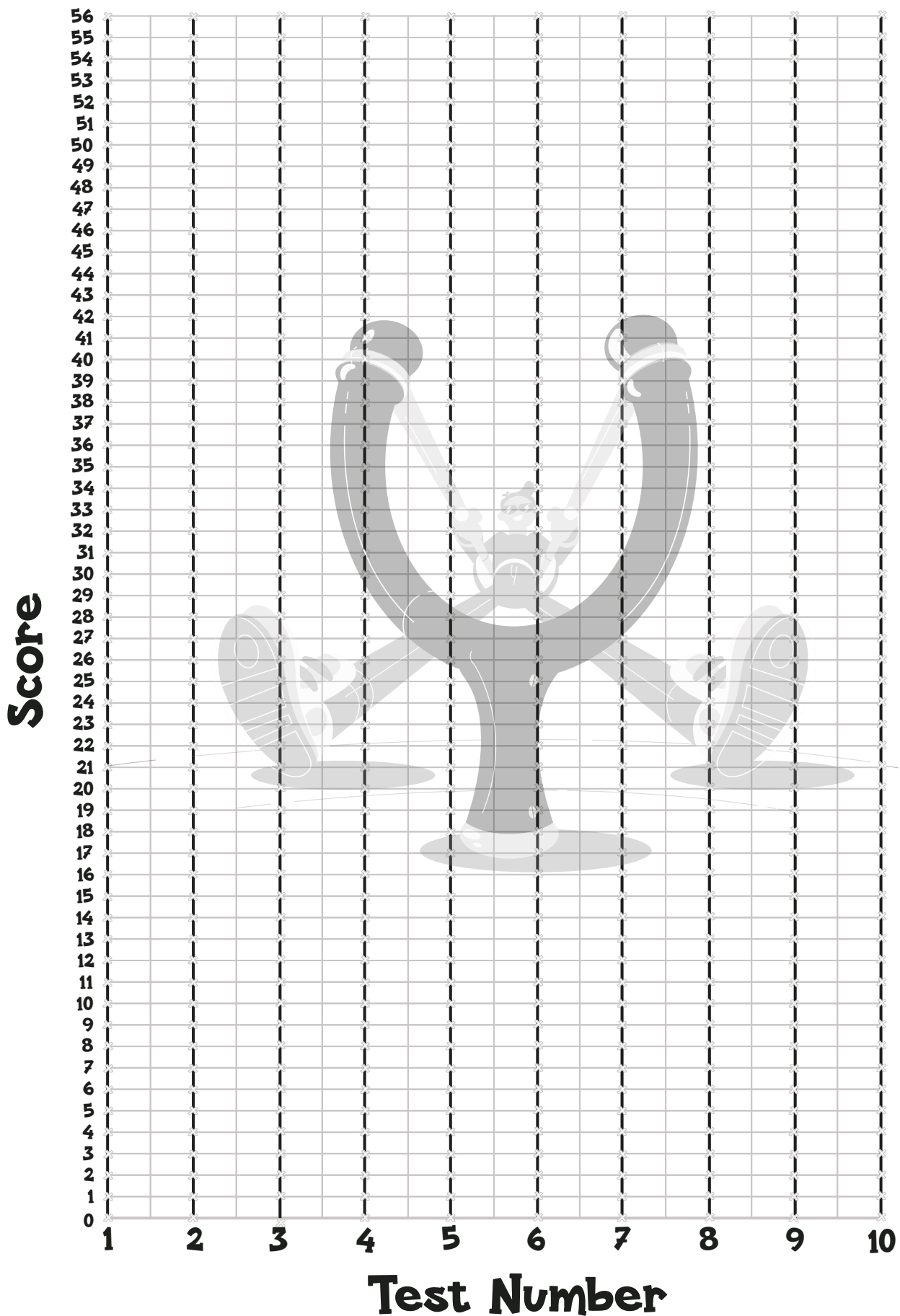
23	tolerably	Most children will be unfamiliar with this adverb. However, the <i>-ly</i> ending is a clue that it is an adverb. The only other obvious adverbs in these lines are <i>oddly</i> and <i>seriously</i> , which narrows our options.
24	on that account	Find three-word phrases on these lines and swap them with <i>for that reason</i> . See lines 7-8: <i>...love you extra on that account [for that reason]...</i>
25	racked her brains	<i>...her thoughts</i> and <i>...her brains...</i> (line 28) seem to match well. This means <i>wrestled with</i> must have a similar meaning, within this context, to racked .
26	quality	See lines 24-25: <i>Bobbie had another quality [characteristic] which you will hear differently described by different people</i> .
27	Ver' good	See lines 31-32: <i>...and 'Ver' good' when they asked him how he had slept</i> .
28	pronoun	Words like I, you, she, them (he, it, we...) are pronouns . They are words that we use <u>instead of nouns</u> (in other words, they are not nouns).

Warm-Up 10 Maths Answers

29	C	50% is one-half. Polly saved one-half of £48, which is £24. Add £48 to £24 to get £72 .
30	E	Connect the points on the grid. Opposite sides will be parallel to each other. All four sides will not be equal in length. There will not be any right angles. It has to be a parallelogram (a 'leaning rectangle').
31	A	It is a triangular-based pyramid , which has four triangular faces.
32	B	Kirsten spent $\frac{3}{4}$ of her £40 in the first hour. 'Divide by the bottom, multiply by the top' to work out how much this is. $£40 \div 4$ then $\times 3 = £15$. She has £25 remaining . $\frac{3}{4}$ of this amount is £15 again ($£25 \div 4$ then $\times 3$). Subtract this from £25 to get £10 .
33	D	A: $(14 \times 10p) + (15 \times 20p) = £4.40$. B: $(20 \times 20p) + (20 \times 10p) = £6.00$. C: $(15 \times 10p) + (17 \times 20p) = £4.90$. D: $(18 \times 20p) + (14 \times 10p) = £5.00$. E: $(4 \times 10p) + (20 \times 20p) = £4.40$.
34	B	The digit 2 is in the hundredths place, which is located two places to the right of the decimal point.
35	C	$60g \times 50 = 3000g$ (6×5 then add two zeros). As there are 1000 grams in one kilogram, this is the same as 3kg .
36	E	2.18m is the same as 218cm (there are 100 centimetres in one metre). The difference between 235cm and 218cm is 17cm .
37	B	There are decimal numbers between 0.6 and 0.7, but we must use <u>hundredths digits</u> . It will take ten hundredths to go from 0.6 to 0.7. Our number is 0.64 .
38	C	The two numbers in the second calculation are each ten times smaller than the numbers in the first one, making the answer <u>100 times smaller than 15480</u> . Alternatively, we could estimate the answer: 86×1.8 will be close to 86×2 . The answer is 154.8 .
39	A	Add each pair of measurements, then double to find each perimeter. However, doubling isn't really necessary in this question. <u>Whichever pair of numbers add to make the largest sum will make the greatest perimeter!</u> It is Rectangle A .
40	B	The product is the answer to a multiplication. $\square \times 9 = 738$ Divide 738 by 9 to find the missing number: 82 .
41	A	We can tell that Driver 2 did not finish the race, as they only travelled 175km - not the required 250km. Driver 1 reached the top of the line graph before the other two drivers.
42	D	A cube has 12 edges (four on the front face, four on the back face, and four edges to connect the front to the back).
43	D	Each wooden chess set costs £80 ($£145 - £65$). $(£145 \times 2) + (£80 \times 2) = £290 + £160 =$ £450
44	C	A reflex angle is greater than 180° but less than 360° . This will be the <u>larger</u> angle between the hands (going <u>anticlockwise</u> from 12 to 5). This covers <u>seven numbers on the clock face</u> ($12 - 5$). There are 30° between each number. $30^\circ \times 7 =$ 210° .
45	D	First, children must know that each angle in an equilateral triangle is 60° . Each corner of the hexagon consists of <u>two</u> of these 60° angles (making 120°). There are six corners, so $120^\circ \times 6 =$ 720° .
46	E	The sequence is $\times 2$, then $\times 3$, then $\times 4$, etc. $600 \times 6 =$ 3600
47	D	$\frac{7}{4}$ can be simplified to become $\frac{1}{2}$, which is 0.5 as a decimal.
48	A	Subtract 7 hours 25 minutes to get back to midnight. Subtract another 1 hour 10 minutes to get back to 10.50pm. We have subtracted 8 hours 35 minutes altogether.
49	C	To find the mean, add the values then divide the total by how many values there were. $46kg + 52kg + 61kg = 159kg$ $159kg \div 3 =$ 53kg
50	A	There are 24 shapes in the container altogether, nine of which are triangles. $\frac{7}{24}$ can be simplified to make $\frac{7}{8}$.

51	9 times greater	Children must be careful. Although each side measurement of Rectangle B is three times greater, its area is <u>nine</u> times greater than Rectangle A's area. $3cm \times 3cm = 9cm^2$. $9cm \times 9cm =$ $81cm^2$.
52	122°	Two angles on a straight line must have measurements that add to make 180° . $180^\circ - 58^\circ =$ 122° .
53	Monday	There are 16 days from Mia's birthday to the end of June, then a further 10 days to reach Lily's birthday. This makes 26 days (3 weeks <u>5 days</u>). 5 days after Wednesday is Monday .
54	5	33 is divisible by 3 and 11. 35 is divisible by 5 and 7. 39 is divisible by 3 and 13. 45 is divisible by 5 and 9. The remaining five numbers (29, 31, 37, 41 and 43) are prime.
55	90.75	Use inverse (opposite) calculations, going from right to left. $528 \div 6 = 88$ $88 + 2.75 =$ 90.75
56	62 cubes	Visualise a complete $4 \times 4 \times 4$ arrangement. This would contain 64 small cubes. Now remove the two small cubes from the front layer to make 62 cubes .

Line Graph of 'Warm Up' Scores



The Catapultalogue



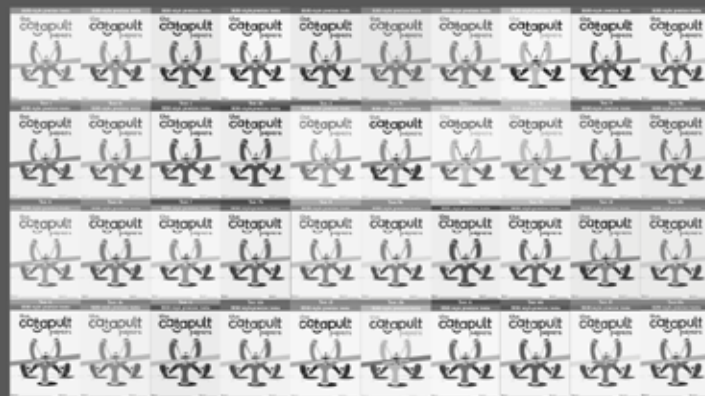
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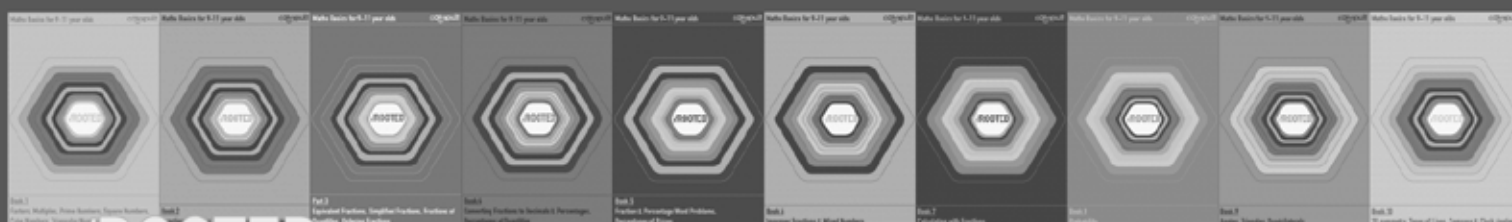
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Catapult is a one-man tuition service. Based in East Belfast, Trevor Veale has been working with primary pupils from 4 to 11-years-old on a full-time basis since 2019.

A former Year 3 and Year 6 teacher, Trevor is now fully booked as a private tutor through to the end of 2030. His teaching resources not only include *The Catapult Papers*, but also *Target Practice* and *Rooted*. All of these materials can be purchased at www.catapult-tuition.com